



“Review and perspective on additive manufacturing of refractory high entropy alloys”

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ABSTRACT

The development of Refractory High Entropy Alloys (RHEAs) for Additive Manufacturing has been advancing rapidly in recent years, buoyed by the triumph of RHEAs generated using traditional techniques in various applications. RHEAs, a recent category of materials possessing extraordinary high-temperature attributes, have only lately emerged. However, owing to their intricate microstructure, the production of RHEAs employing conventional methods such as casting and forging poses a challenge. On the other hand, Additive Manufacturing (AM) has exhibited significant potential for creating RHEAs due to its ability to generate intricate microstructures and complex geometries with minimal material wastage. The fundamentals of AM and RHEAs are introduced at the outset, in conjunction with the unique attributes of 3D-printed RHEA products, such as their enhanced strength and unique microstructure. The techniques for producing RHEA powders, including atomization and mechanical alloying, as well as the characteristics of the powder, are then elucidated. In this review article, we will provide a comprehensive overview of RHEAs produced using Additive Manufacturing (AM). We will examine the most innovative AM methods presently available for producing RHEAs, their microstructure, attributes, and potential applications.

1. Introduction

The emergence of RHEAs is intricately connected to two elements, precisely, the refractory metals and the high entropy alloys (HEAs). The concepts of "high-entropy alloy" and "multi-principal alloy," which upended the design of conventional alloys and reinvented the process of designing alloys, were initially put forth in 2004 by Yeh and Cantor et al. [1,2]. The majority of HEAs are made up of five or more different types of elements. Additionally, the quaternary alloy is sometimes denoted as HEAs, wherein the elements are skillfully merged with atomic ratios that are either identical or strikingly similar [3,4]. Refractory metal alloys possess a broad spectrum of prospective applications due to their elevated density, exceptional thermal conductivity and high-temperature capabilities, remarkable robustness and toughness, and minimal sputtering yield and hydrogen interactions. Illustrative examples encompass plasma-facing components designed for fusion reactors, fusion targets, armor penetrators, and nuclear space power and propulsion. The utilization of structural alloys, specifically the RHEAs, renowned for their elevated melting points, is highly advantageous in demanding situations. The term 'refractory metals' has been defined in

various ways, but an exceptional melting point is a common requirement for their inclusion. Some definitions specify a melting point above 4000 °F (2200 °C) as a necessary condition for qualification. The five elements, namely niobium, molybdenum, tantalum, tungsten, and rhenium, are included in all definitions. However, a broader definition includes nine more elements with a melting point above 2123 K (1850 °C), namely titanium, vanadium, chromium, zirconium, hafnium, ruthenium, rhodium, osmium, and iridium [5,6].

Additive Manufacturing (AM), in contrast to conventional production methods such as arc melting [7–9], induction melting, and mechanical alloying, offers unparalleled freedom in geometric design and accelerated prototyping capabilities across numerous applications [10–13]. In contrast to standard metal complexes, high entropy can help promote a unique phase structure and significant lattice deformation (strain). These substances consist of several metallic elements in proportions that are either equal or very close to equal, and they possess a structure of a disordered solid solution with an increased configurational entropy [14–16].

RHEAs offer a wide range of possible applications in addition to their remarkable qualities, such as aerospace and defense, energy production

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and storage, and biomedical applications [17–19]. The utilization of additive manufacturing in RHEA production offers several benefits, including the ability to construct complex geometries with high precision and accuracy with minimal material waste and create gradient structures that can enhance RHEAs' inherent qualities. In addition, RHEAs produced using AM have the potential to transform the field of advanced materials entirely and make it possible to manufacture complex components with exceptional qualities. RHEAs are especially suitable for use in demanding conditions due to their excellent resistance to corrosion and wear and their ability to maintain optimal performance even at high temperatures [20].

In recent years, the benefits of additive manufacturing (AM) technology have received considerable attention. AM offers numerous advantages, including reduced flaws, high precision, high efficiency, and the ability to produce complex-shaped parts with ease [21–23]. The beneficial aspect of additive manufacturing (AM) for RHEAs resides in its remarkable ability to transcend the boundaries of conventional manufacturing methods, like casting and forging, that frequently encounter restrictions imposed by the heightened melting points and subpar malleability of RHEAs. Metal powders for AM have been created using a variety of techniques, including gas atomization [24], water atomization [25], plasma atomization [26], mechanical alloying [27], chemical reduction, carbonyl process, plasma spheroidization [28], need to be more in symmetrical according to AM specifications.

AM for RHEAs presents several limitations that must be considered despite its advantages. Among the most notable is the necessity for specialized equipment and expertise, which may prove costly and time-intensive. Moreover, the size of the components that can be produced and the accuracy and uniformity of the result can pose significant challenges for AM methodologies [29–33] [29–33] [29–33]. This article aims to provide an in-depth and comprehensive analysis of the additive manufacturing of RHEAs through an examination of powder development, printing techniques, microstructures, product properties, and potential applications.

2. Development processes of RHEA powders for AM

Currently, RHEA powders are created by mechanical alloying [34], gas/water atomization, and Plasma rotating electrode techniques. Atomization and Plasma rotating electrodes can make spherical powder particles. In contrast, mechanical alloying creates irregular-shaped powder particles that must be treated to reach high levels of sphericity for optimal printability. The process parameters in AM are also influenced by powders' thermal parameters (specific heat, thermal conductivity, latent heat, and expansion coefficient) and optical parameters (absorption, reflection, and transmission). Therefore, creating printable RHEA powders requires a thorough understanding of such features.

2.1. Mechanical alloying

In a high-energy ball mill, particles are repeatedly cold welded and fractured as part of the mechanical alloying (MA) process. This method is frequently used to make RHEA powders and allows for the creation of uniform powder combinations. The procedure typically entails mixing the appropriate components in the correct ratios and milling the mixture for a predetermined time to produce a uniform powder.

According to Fig. 2, elemental powders split and fuse during the milling process to create RHEA powder particles as a result of the spinning of the grinding bowl (see Fig. 1) [35,36]. Extreme hydrostatic compression causes elemental powder particles to be trapped between colliding grinding balls and undergo significant plastic deformation before shattering. Table 1 and Table S1 summarize the different RHEAs with different processing parameters for making mechanical powders.

The milling procedure involves repeated cold welding and fracture, which results in ongoing particle refining. Powder particles with a steady-state size distribution and evenly distributed components are produced when the repetitions of welding and fracturing are balanced (Fig. 2). The mechanical alloying process offers several key advantages, including an improved ability to dissolve solids, process materials at room temperature, and a uniform distribution of elements. A critical

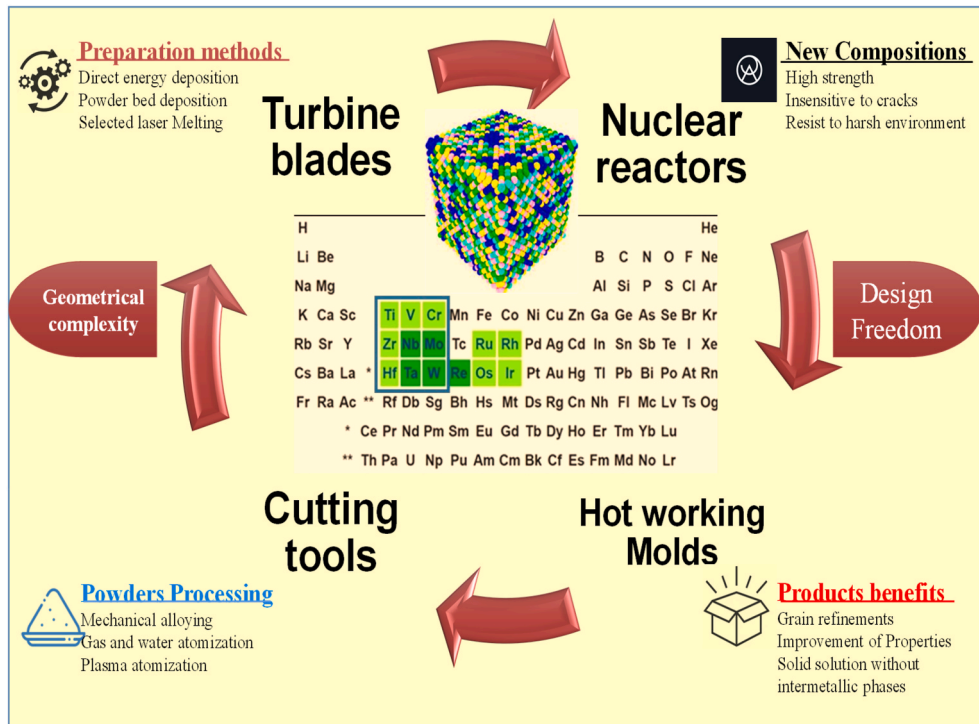


Fig. 1. Overview of the relationship between RHEAs and Additive manufacturing processes concerning the Applications, powder development, product characteristics, and potential applications.

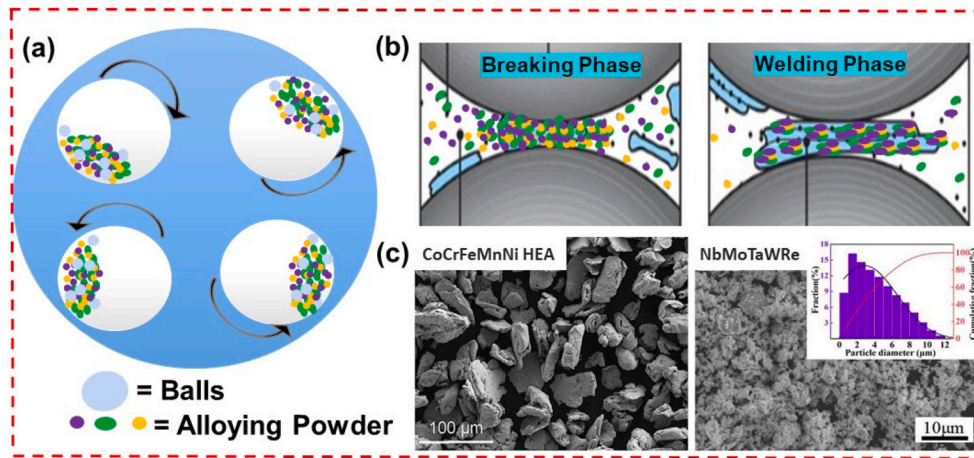


Fig. 2. Systematic illustration(a),(b) of ball milling experiment for making powders,(c) Cryogenic ball-milling by a 0.5 h planetary ball-milling at room temperature of CoCrFeMnNi [24], the morphology of 30 h-milled powders and the size distribution of powder particles of NbMoTaWRe [37].

Table 1

RHEA powders developed by mechanical alloying about the feedstock, milling parameters, and phases.

RHEAs	Phase	Crystallite size [nm]	Lattice Strain [%]	Purity [wt %]	Particle size [μm]	Milling parameters			Apparatus	Ref
						Ball-to-powder ratio	Time [h]	Speed [rpm]		
MoNbTaTiV	BCC	12.9	0.89	99.9	–	10:1	40	350	QM-3SP4	[38]
WNbMoTaV	BCC	66.1	0.96	99	66.1	10:1	6	300	Pulverisette	[39]
Al _{0.3} NbTa _{0.8} Ti _{1.5} V _{0.2} Zr	BCC + HCP	–	–	99.7	–	10:1	50	250	Pulverisette 6	[40]
Ti _x NbMoTaW	BCC	10.8	–	99.6	–	15:1	12, 26, 45, 60	400	QM-DY4	[41]
AlCoCrCuFeTi	FCC + BCC	13–58	0.096–0.231	–	45	10:1	20	300	Pulverisette-P5	[42]
Al _{0.4} CoCr _{0.5} FeNiTi _{0.6}	FCC + BCC	7	0.175	99.6	30	10:1	60	–	QM-3SP4	[43]
Al _x CoFeNi _{1.5} V _{0.5} Cu _{1-x}	FCC + BCC	6.3–7.7	1.21–1.52	99.5	48	10:1	70	300	QM-3SP4	[43]
AlCo _x FeNiSiB	BCC	–	–	99.5	75	15:1	140	350	–	[44]
Al _{0.4} CoFeNiSi _{0.4}	FCC + BCC	–	–	99	–	20:1	70	300	QM-QX4	[45]
AlCoNiZnCu	FCC	14.5	0.84	99.5–99.8	60	10:1	50	300	Pulverisette-P7	[46]
Al _{0.15} CoCrCuFeNiTi _x C	FCC + BCC	–	–	99.5–99.5	180	15:1	36	500	KMC-1BV	[47]
AlCoCrFeNiSi	FCC + BCC	11.7	0.73	99.5	75	10:1	30	300	QM-3SP2	[48]
AlCoFeNiCu	FCC + BCC	7.1	–	99	–	5:1	10	–	SPEX-8000 M	[49]

factor in alloying is the milling time, which significantly impacts the phases, crystal sizes, and lattice strains of the RHEA powder particles. Through the repeated process of cold welding and crushing, the mechanical alloying process promotes the dissolution of different constituent elements and the formation of solid solutions. Elements with low melting points or brittle crystal structures tend to dissolve more. Like atomization, the phases of RHEA powders produced through mechanical alloying depend on their constituent elements and proportions. Certain elemental phases remain even after the alloying process in some powder particles. This can be attributed to the fact that the solid solubility of these elements is lower than the nominal composition. A powder matrix with high storage energy and a large free volume can be used to extend the solid solubility of the elements. The high storage energy provides a robust thermodynamic driving force for atomic migration, while the sizeable free volume creates numerous diffusion channels for solid-state alloying. The milling duration is a critical alloying parameter that affects the phases, crystal sizes, and lattice stresses of RHEA powder particles.

2.2. Atomizations: (gas, water, plasma)

In recent years, atomization has emerged as a promising method for creating pre-alloyed powders for DED and PBF. It uses high pressure from gas, water, plasma, or rotational forces to split a narrow stream of liquid metal droplets into spherical powder particles after solidifying. It is said that printable RHEA powders have been created using water and gas atomization. The most widely used technique for producing fine metal spherical powders for additive manufacturing is gas atomization [35,50,51]. Figure 04 depicts a diagrammatic representation of the experimental configuration of diverse atomizations. RHEA ingots or uncontaminated constituents of compositions necessitate prior melting in a crucible melting furnace to initiate the formation of a liquefied stream. A high-pressure, high-speed gas is subsequently employed to make contact with an atomizer, fragmenting the stream into droplets. During this process, the conversion of the kinetic energy of the gas into the surface energy and kinetic energy of the droplets occurs through the genesis of a shockwave in the gas/melt collision zone. Preceding the

droplets' impact on the walls of the atomization chamber, they condense into powder particles while suspended in the air. The stream disintegration mechanism comprises two stages [52]. In the first phase, the sinusoidal oscillations of the high-pressure gas disrupt the molten metal, leading to the formation of large droplets or unstable ligaments due to surface tension and fluctuations. In the subsequent stage, the droplets or ligaments undergo further stripping. They are converted into spherical powder particles of varying sizes in motion through either a low-turbulence or a more disorderly high-turbulence method (refer to figure 04) [53,54]. By optimizing the gas pressure, gas flow rate, and melt flow rate in gas atomization, RHEA powders with high levels of sphericity, uniform distributions of their constituent elements, and acceptable particle sizes can be created [55,56]. The powder flowability will be negatively impacted by satellite particles produced by the solidification of tiny molten droplets attached to the surfaces of big droplets [57]. A thorough process optimization is needed to reduce these satellite particles and enhance the powder flow [55].

Particle size significantly impacts the morphologies and phases of RHEA powders. Ding et al. investigated effects of arc melting method of gas atomized powder of $\text{AlCoCrFeNi}_{2.1}$ alloy [55]. The powder has soft magnetic material properties and superior corrosion resistance to stainless steel powders. This work shows that the $\text{AlCoCrFeNi}_{2.1}$ alloy in bulk generated by the arc-melting process has a crystal structure different from that of powder as atomized, which is an FCC and BCC dual-phase. Zhao et al. found in another work that gas-atomized $\text{Al}_{0.6}\text{CoCrFeNi}$ powder particles larger than 75 nm were made up of FCC + BCC phases. In comparison, particles smaller than 75 nm only showed a BCC phase. The primary cause for this occurrence is predominantly brought about by the eutectic process undergoing suppression due to an elevated cooling velocity [57]. In addition to preventing eutectic solidification, the rapid cooling in the tiny particles made Al more soluble in CoCrFeNi, forming the BCC phase. Additionally, the

reduction in particle size enhances the RHEA powder particles' surface quality.

Similar in nature to the gas atomization process, water atomization involves the utilization of water jets as the atomizing agent instead of high-pressure gases. The water-atomized powders are typically subjected to a series of thermal treatments involving heating, dehydration, and drying [58,36]. Due to the cooling medium, water, having a higher specific heat capacity, the cooling rate for water atomization is 10–100 times greater than that of gas atomization. While molten droplets with minor surface tensions are more likely to form irregularly shaped particles, droplets with immense surface tensions will solidify into nearly spherical powder particles. This explains why HEA powder particles atomized in water often have less sphericity than those atomized in gas (Fig. 3(b)). While the plasma atomization process was carried out in an argon-inert atmosphere, the gas atomization process used high-purity argon as the atomizing and filling gas. In addition, as shown in Fig. 4 (c), the plasma-atomized powders have significantly greater surface smoothness and particle sphericity than the Gas and water-atomized powders. Particles were sieved into one batch individually and kept within a size range below 150 μm , where no pores could be discovered [59].

3. Additive manufacturing techniques for RHEAs

Due to the unique ability of the technology to print products with design freedom and geometric complexity, additive manufacturing of RHEAs has garnered increasing interest from academia and industries. The utilization of additive manufacturing techniques, namely DLD, Laser Cladding, PBF, DED, and SLM, has been extensively analyzed for the production of RHEA products. These aforementioned methods employ a focused, high-energy beam to interact with powders, forming a melt pool followed by rapid melting and solidification. This swift

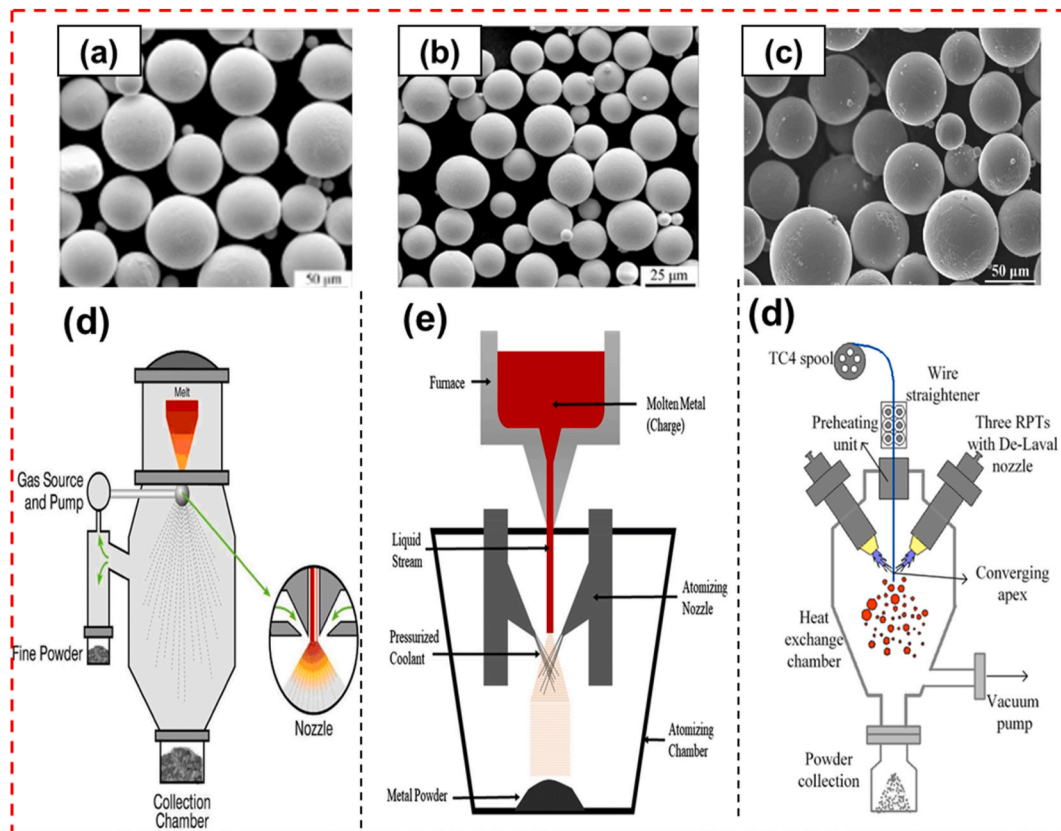


Fig. 3. Morphologies of (a) Gas-atomized [57], (b) water-atomized [60], and (c) plasma-atomized powders [59], Illustration of a typical experimental setup for (d) gas atomization [61], (e) Water atomization, (f) Plasma atomization [26] and its schematic mechanism of atomization stages.

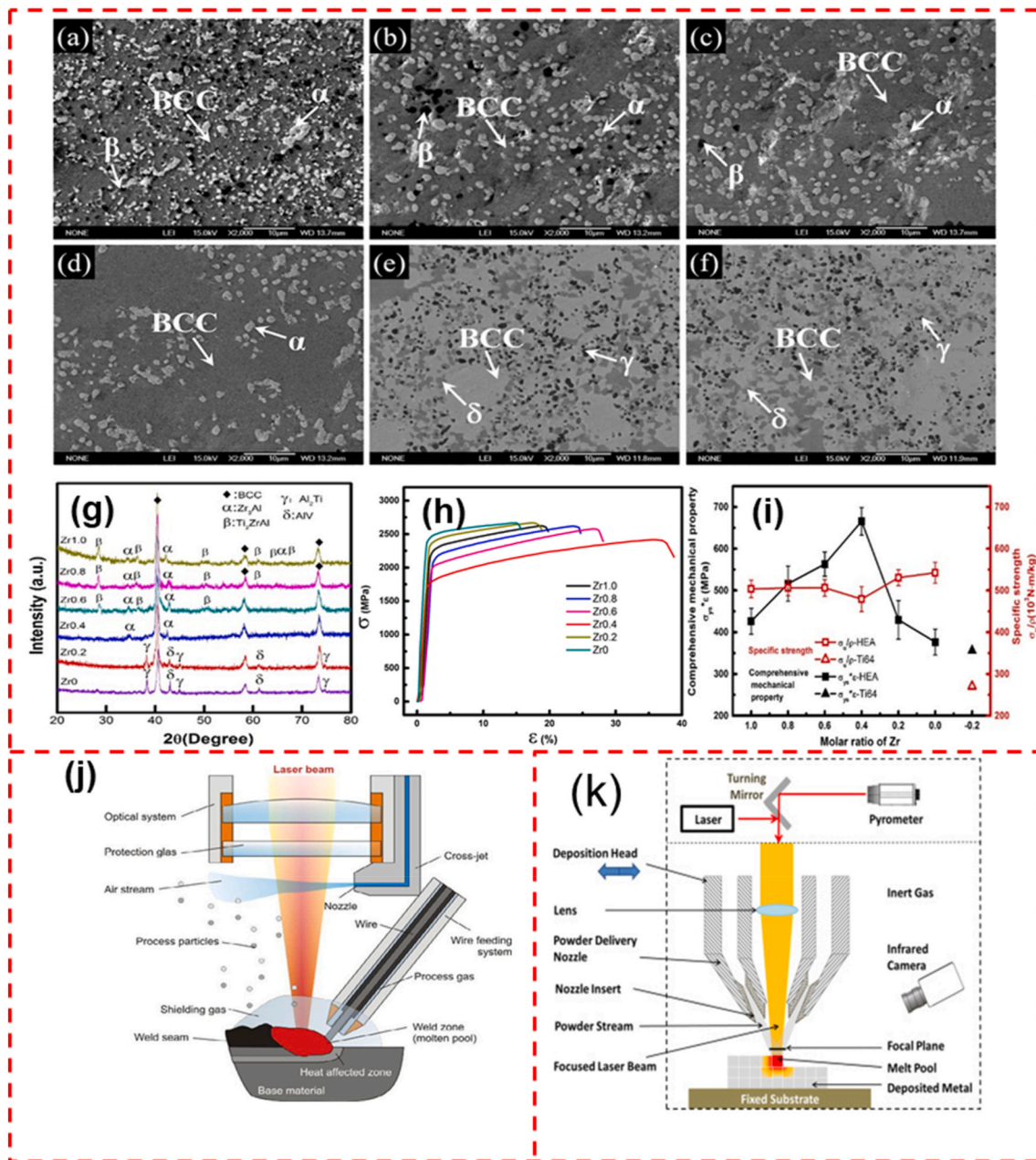


Fig. 4. Microstructures of the $\text{Al}_2\text{NbTi}_3\text{V}_2\text{Zr}_x$ alloy: (a) Zr1.0; (b) Zr0.8; (c) Zr0.6; (d) Zr0.4; (e) Zr0.2; (f) Zr0, (h) XRD patterns of the $\text{Al}_2\text{NbTi}_3\text{V}_2\text{Zr}_x$ bulks sintered at 1250 °C., (i) Stress-strain curves obtained during the compression tests of the $\text{Al}_2\text{NbTi}_3\text{V}_2\text{Zr}_x$ alloy, (g) Effect of Zr molar ratio on the comprehensive mechanical properties and specific yield strength of $\text{Al}_2\text{NbTi}_3\text{V}_2\text{Zr}_x$ alloys [62] (j) Illustration of Laser cladding [63], (k) Blown powder Direct Laser Deposition (DLD) with thermal monitoring [64].

solidification process effectively hinders elemental segregation that typically arises at linear defects, such as dislocations, or surface defects, like stacking faults, grain boundaries, phase boundaries, etc. Additionally, the formation of brittle intermetallic compounds is prevented, thereby augmenting the mechanical properties of the products. The advantages and disadvantages concerning the utilization of PBF and DED for the fabrication of RHEAs shall be deliberated in this section.

3.1. Direct laser deposition (DLD)

Direct laser deposition (DLD) and electron beam melting (EBM) are the additive manufacturing processes most frequently utilized to create RHEAs. DLD makes it possible to create RHEAs out of elemental powders and simplifies their compositions as the process progresses [65].

However, compared to DLD, SLM and EBM have significantly smaller beam diameters and are more able to fine-tune the microstructure. Joseph et al. [66] successfully created three alloys based on the Al_x-CoCrFeNi HEA system with varying molar fractions of Al ($x = 0.3, 0.6$, and 0.85) using DLD. Similarly to this, Sistla et al. [67] study on the effects of the Al/Ni ratio on the microstructure and phase development in the DLD Al_xFeCoCrNi_{2-x} HEA. It was discovered that the fabricated RHEA's microstructure and phase development were significantly impacted by the Al/Ni ratio. Additionally, effort has been made to create MoNbTaW RHEAs from an equimolar mixture of elemental powders utilizing the DLD approach [68]. This study demonstrated that it is possible to create RHEAs utilizing pre-mixed powders and in situ alloying during the DLD process. The concentration of an element would be higher and lower its melting point, and it was discovered that the

chemical composition deviated from the composition of the beginning powder. More importantly, the RHEA composition can be customized by carefully adjusting the processing parameters. An additional remelting procedure was required after the powder deposition steps to fabricate RHEAs with elements with various melting points successfully. The compositional homogeneity should be carefully taken into account when introducing remelting. The manufacture of RHEAs may also be hampered by oxygen in the chamber because all refractory metals are susceptible to oxidation [69]. The hardness of another DLD-manufactured RHEA AlCrFeNiTa can reach 860 Hv, which is three times higher than arc-melted AlCoCrFeNi HEA specimens because of the equiaxed dendritic microstructure containing two phases. Ocelik et al. discovered that the hardness of DLDed AlCoCrFeNi HEAs was around 492 Hv due to the formation of the BCC phase [70].

Due to the high melting points of refractory elements, high temperatures were frequently used in the arc melting process to create lightweight RHEAs, and as a result, many of these RHEAs have high purity levels. Due to the complexity of the composition and wide variations in elemental melting temperatures during the smelting process, micro-segregation also exists, which leads to the production of dendritic and interdendritic microstructure. As can be observed in Fig. 4., the BCC matrix for Zr_{1.0} alloy has a significant quantity of brilliant block and dark secondary phases distributed as granules. When the Zr concentration is reduced from $x = 1.0$ to $x = 0.4$, the volume fractions of these secondary phases also decrease (Fig. 4(b–d)). In Zr_{0.4} alloy, the black phase disappears, and a negligibly small amount of the light phase appears. According to the combined XRD and SEM data, the dark phase is (Ti₂ZrAl), and the bright block precipitate is (Zr₃Al). When the Zr concentration is further reduced to less than 0.2, the phase completely vanishes, but a significant number of new dark and light grey phases, as seen in Fig. 4(e and f), form instead. All of the Al₂NbTi₃V₂Zr_x alloys have excellent yield and fracture strengths, and some plasticity is shown in Fig. 4(h and i). With a drop in Zr concentration, the alloy's yield strength and fracture strength are reduced while its fluidity improves. This is notably true for the Zr_{0.4} alloy, where the compressive strain approaches 38.2 %. However, the strength of the alloy increases, and the compressive strain falls to 18.5 % and 15.7 %, respectively, as Zr concentration is further reduced to 0.2 and 0 progressively. The quantity of secondary phases is likely a factor in this phenomenon. The overall mechanical property of Al₂NbTi₃V₂Zr_x alloys increases initially and then falls when the Zr molar ratio lowers. The most superior comprehensive mechanical property of 665.4 MPa was demonstrated by the Zr_{0.4} alloy (YS = 1742 MPa, $\epsilon = 38.2\%$), which is about twice that of Ti₆₄ (356 MPa) [62].

3.2. Laser cladding

Laser cladding (LC) can create a variety of surface alloys and composites with the necessary characteristics. Using laser beam cladding in surface engineering makes it possible to make solid surface coatings with evenly dispersed hard particles in softer and tougher matrices free of porosity and fissures. Several methods, such as powder blowing, wire feeding, pre-applied powder coating, etc., feed the material into the substrate surface underneath the laser beam during the process shown in Fig. 4(j). Because it can create intricate and high-quality parts, laser cladding is a popular additive manufacturing (AM) technique for RHEAs. The laser cladding of TiC-reinforced CoCrFeNi high entropy alloy coatings on Ti–6Al–4V substrates is the subject of a study by Yuan et al. [71]. The laser-cladding coatings, which the researchers discovered to have exceptional wear and corrosion resistance, are suitable for various industrial applications. AlCoCrFeNi high-entropy alloy coatings were laser-clad onto Ti–6Al–4V substrates in the study by Y. Wang et al. [72]. The researchers discovered that the laser-cladding coatings showed better high-temperature oxidation resistance, making them suited for high-temperature applications. The microstructure and characteristics of laser-clad AlCoCrFeNi_{2.1} eutectic high-entropy alloy coatings on Ti–6Al–4V are studied by Y. Zou et al. [73]. The scientists

discovered that the laser-cladding coatings had excellent wear resistance, corrosion resistance, and hardness, making them appropriate for various industrial applications. TiC/AlCoCrFeNi high-entropy alloy coatings were laser-clad onto Ti–6Al–4V substrates, according to a study by C. Wang et al. [72].

The determination of quality and performance in DED and PBF products relies upon the attributes of the powder utilized. Inadequate powder quality can instigate deficiencies in printed items, including but not limited to porosity, cracks, inclusions, and suboptimal surface roughness. It has been suggested that RHEA powder particles possess high mixing entropies exceeding 1.5R. RHEA powders are manufactured using mechanical alloying, gas atomization, and water atomization techniques. Spherical powder particles can be generated through gas and water atomization, whereas mechanical alloying engenders powder particles of irregular shape that necessitate treatment to attain elevated levels of sphericity for optimal printability. To enhance the printability of these powders, a diverse range of post-processing techniques and treatments can be utilized. Several prevailing methods include sieving, screening, sintering, surface coating, post-annealing, powder blending, and chemical treatment.

3.3. Powder bed fusion (PBF)

One of the most popular additive manufacturing (AM) processes for creating RHEAs is powder bed fusion (PBF). Powder bed fusion (PBF) methods primarily come in two types: selective laser melting (SLM) and electron beam melting (EBM). In the SLM method, a laser beam uses a 3D computer model to melt and fuse metal powdered particles layer by layer selectively. The powder bed is heated and melted by the laser radiation, which hardens when the energy source advances to the following layer. PBF is a well-known additive manufacturing method for metals, alloys, and occasionally ceramics. In this method, the powder is put in layers that range in thickness from 20 to 150 μm , and then the targeted area is selectively melted using a laser beam (either a continuous or pulsed wave). Build layer thickness, laser power, scan speed, and hatch spacing are crucial factors affecting how well materials are built. These variables affect the processing conditions' volumetric energy density. Another essential factor that affects how much laser energy is linked to the powder layer is laser absorptivity. Notably, PBF is frequently carried out in an argon environment with oxygen concentrations between tens and hundreds of parts per million (ppm).

3.4. Electron beam melting (EBM)

Fig. 5(e) displays a schematic of the EBM's equipment. EBM melts metal powders using an electron beam [74]. EBM produces components within a 104 mbar or higher environment, characterized by a high-vacuum setting. This is due to the distinctive operating conditions of the electron beam, which create an optimal manufacturing environment free of contamination. Despite operating under different conditions, EBM shares a principle similar to SLM. A unique method used in EBM to lessen the thermal gradient in printed items is to pre-heat the powder bed utilizing an electron beam at a temperature of up to 1100 °C before scanning and melting [75,76].

A new generation of RHEA products has been developed using the CoCrFeNiTi system [79]. A comparison between SLM-printed and EBM-printed Co_{1.5}CrFeNi_{1.5}Ti_{0.5}Mo_{0.1} HEA products with a relative density of 99.4 % was done [80]. Single cubic and FCC phases could be seen in both products, whereas Ni₃Ti brittle intermetallic complexes only precipitated in the EBM-printed one. It was attributed to the significantly slower cooling rates brought on by the preheating and larger melt pool sizes in EBM [81], which gave precipitation enough time to happen. Additionally, the RHEA products produced by EBM and SLM printing differed in columnarization for crystal grains and anisotropy. The higher G/R ratio hampered EBM's development of equiaxed grains [82].

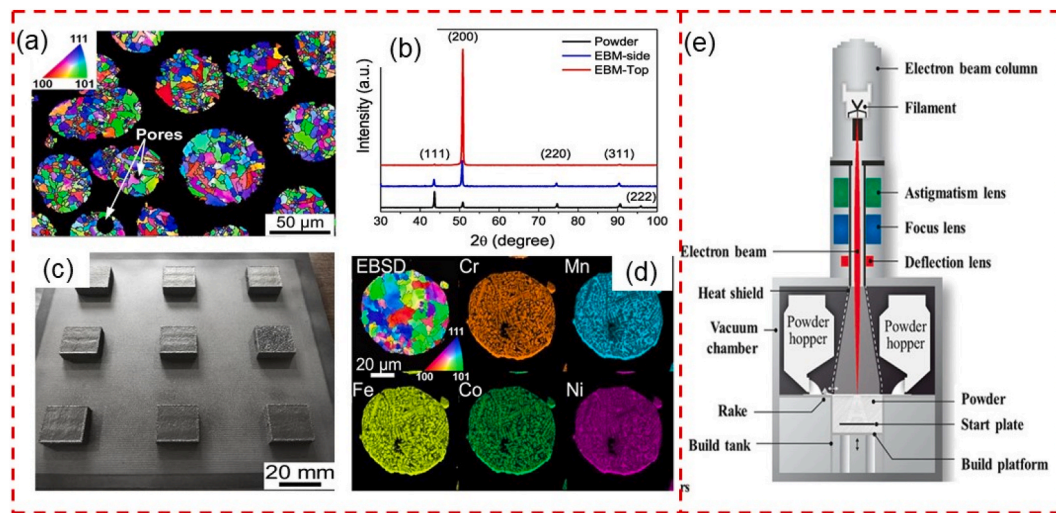


Fig. 5. (a)EBSD showing the polycrystalline of HEA powder and (b) XRD patterns of the powder and the additively manufactured part [77]. (c) Image of EBM-built part, (d) EBSD image and the corresponding elemental distribution of the as-received powder [77], (e) Schematic diagrams of major 3D printing processes for HEA products [78].

Wang et al. executed an optimization procedure for electron beam melting (EBM) in the production of CoCrFeMnNi goods, employing a gas-atomized high-entropy alloy (HEA) powder [77]. The HEAs under consideration exhibited columnar grains having intragranular cellular grains along the build direction with a strong $\langle 100 \rangle$ texture, resulting in a relative density of 99.4 %. The columnar grains were a few millimeters in length, with an average width of 65 μm , while the cellular grains obtained an average width of 2.9 μm (Fig. 5(a–b)). Mn and Ni elements are segregated into interdendrites, whereas Fe, Cr, and Co elements are segregated into dendrites (Fig. 5(d)). It should be noted that reproducible production and consistent metallurgical bonding depended on surface quality. By optimizing the process, the required level and smooth surfaces of printed HEAs could be achieved without swelling and pores. The enlarged top surface was the upshot of ferocious melt pool motion brought forth by temperature gradients, Marangoni convection, and element vaporization (Fig. 5(a)). The aforementioned observation implies that a considerable melt pool and elevated melting temperatures promoted the generation of surface swelling and irregularities. To sustain identical melting conditions with equivalent energy inputs for RHEA products that are fully melted and feature homogeneous top surfaces, it may prove beneficial to manipulate various parameters, including but not limited to line offset, focus offset, scanning speed, reference current, and preheating temperature.

V.V. Popov et al. reported the formability of $\text{Al}_{0.5}\text{CrMoNbTa}_{0.5}$ RHEA by EBM utilizing elemental powder mixes [83]. It is possible to dispense with the pre-heated base plate, pre-sintering, layer-by-layer melting, and final removal from the base plate. Recent research uses pre-alloyed WMoTaNb powder and pure Ti powder to create EBM-produced WMoTaNbTi RHEA [84]. It was discovered that only one bcc phase formed even though significant Nb and Ti vaporization took place throughout the EBM processes, leading to a high strength of 1312 MPa. The correct inclusion of Ti successfully reduced the ripping effect caused by residual stress after solidification and effectively corrected local cracking caused by thermal shrinkage during solidification. As a result, the product's good integrity was preserved, and the likelihood of crack formation significantly decreased. TaNbMo-based and (TaMoNbCr)Al solid solutions were formed during the in situ alloying of blended elemental powders using EBM, according to research on $\text{Al}_{0.5}\text{CrMoNbTa}_{0.5}$ HEA products; however, further study is still needed to fully understand the microstructure homogeneity and mechanical properties of the RHEA [83]. However, the main issues continue to be the unavoidable porosities, residual stresses, and the presence of

non-alloyed powders. Several papers [14,79,85–88] have already been published discussing the potential benefits of EBM for the production of HEAs, but all of them discuss experiments with pre-alloyed powders. In general, the higher beam power in the EBM compared to the laser-based PB AM can make the loss of the alloying elements from uneven evaporation more substantial.

3.5. Selective laser melting (SLM)

Zhang et al. [89] fabricated WTaMoNb RHEA from the SLM method experimentally and optimized the parameters for the SLM process. The warping and cracking problems were produced by the uneven temperature distribution over the entire part, according to the thermal-mechanical study carried out by simulation. Based on the thermal-mechanical study and simulated trials, the procedure was further enhanced. The trial using the revised method was carried out, and its efficacy for producing alloys with any number of layers without breaking flaws was confirmed. As shown in Fig. 6(f), the manufactured portion had a flat surface with a height greater than 0.5 mm. The entire SLM process might be simulated and improved with the help of the thermal and mechanical models in combination with the FD-FE technique. Due to the different heat conductivity coefficients between the powder and the HEA solid, as shown in Fig. 6(c), the temperature showed an uneven distribution at the surface. However, as the observation layer depth grew, the degree of asymmetry decreased.

With a BCC phase and a small amount of the HCP phase by SLM, Liu et al. [90] successfully prepared the WMoTaTi RHEA for the most recent study. The microhardness was 621 HV, which is higher than the standard preparation method. This investigation demonstrates the strong overall mechanical qualities of the SLM-manufactured WMoTaTi RHEA. There are further additive manufacturing techniques that can be used to create RHEAs. For example, Cui et al. [90] used a laser cladding technique to create a gradient FeCoCrNiMnAl HEA coating, displaying an excellent combination of strength and toughness. By combining mechanical mixing and ball milling to create elemental powders, Peng et al. [91] created a new $\text{V}_{0.5}\text{Nb}_{0.5}\text{ZrTi}$ RHEA via SLM. In-depth research has been done on the mechanism of the SLMed $\text{V}_{0.5}\text{Nb}_{0.5}\text{ZrTi}$ RHEA crack formation. Consequently, a small process window is required to achieve high forming capability and strength for SLM.

The limitations of SLM are derived from the size of the components that can be produced, particularly for larger RHEA components. The quality and consistency of the final product can be compromised by

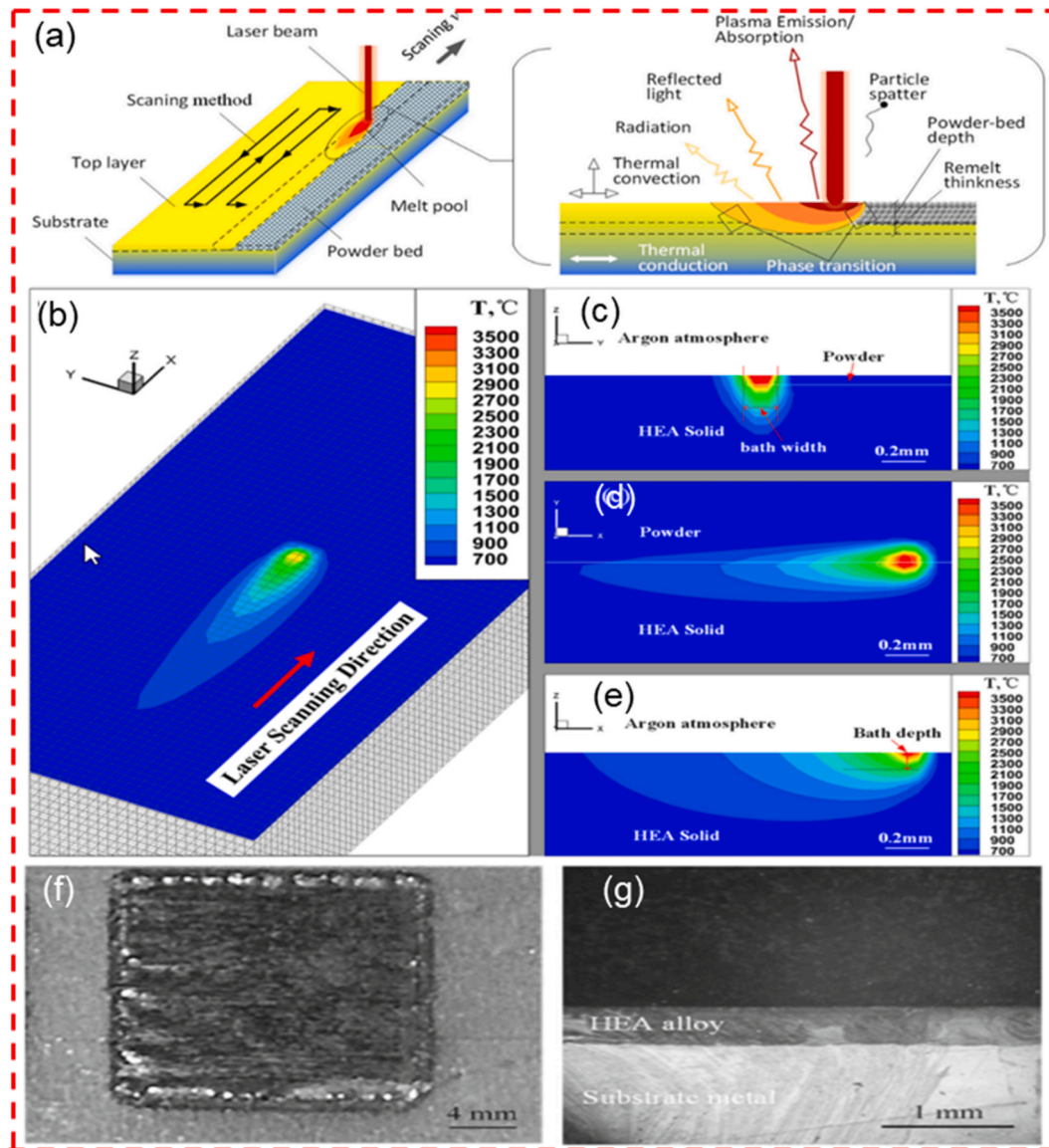


Fig. 6. (a) Schematic of thermal behaviors during the SLM scanning, Temperature distribution in microzone ($4\text{ mm} \times 2\text{ mm} \times 1\text{ mm}$) around laser spot for SLM process in 200 s. (b) 3D view. (c) Top view. (d) Longitudinal section perpendicular to the y-axis (e) Longitudinal section perpendicular to the x-axis (f) Sample produced via SLM process with lower height. (g) Vertical section [89].

differences in powder composition, particle size distribution, and flowability, leading to flaws such as porosity and cracking. The acquisition of specialized tools and knowledge required for PBF can be a costly and time-intensive process. SLM may also be hindered by the high melting temperatures of RHEAs, which can result in thermal stress and warping during the printing process. To overcome these challenges and improve the consistency and quality of the final product, researchers have explored a number of methods, such as preheating the powder bed or utilizing multiple lasers [92].

3.6. Directed energy deposition (DED)

DED is another popular AM technique utilized for fabricating RHEAs. This method involves the utilization of a high-energy heat source, such as a laser or electron beam, to selectively melt a feedstock material, which is usually in the form of a wire or powder. The final product is constructed by adding layer upon layer of the melted material in a controlled manner [82,93]. Metals undergo a process of melting subsequent to deposition in the form of powders or wires, utilizing a laser,

electron beam, or electric arc. Several terms, such as laser metal deposition, laser engineered net shaping, laser powder deposition, direct laser deposition, direct metal deposition, and 3D laser cladding, are employed to delineate the DED process based on melted powders. Powders are emitted from nozzles, and a focal point is established for powder flow, as illustrated in Fig. 7. The powder can fuse in a melt pool that flows in tandem with the laser when the laser and powder flow's focal points are aligned for deposition. DED is often carried out in an argon-filled inert atmosphere where the oxygen content can be lowered to 5–10 ppm. In contrast to PBF, DED can print RHEA products using both RHEA powders and various immaculate elemental powders that are supplied simultaneously through numerous hoppers. Through the optimization of the DED process parameters, homogeneous element distributions can be obtained in melt pools created by melting and depositing elemental powders onto a substrate. RHEA melt pools, the foundation for printing RHEA goods, can be created through in situ alloying after the solidification. The qualities of the printed goods are greatly influenced by the process parameters in DED, such as laser power, scanning speed, powder flow rate, and hatch space. The phases and crystal

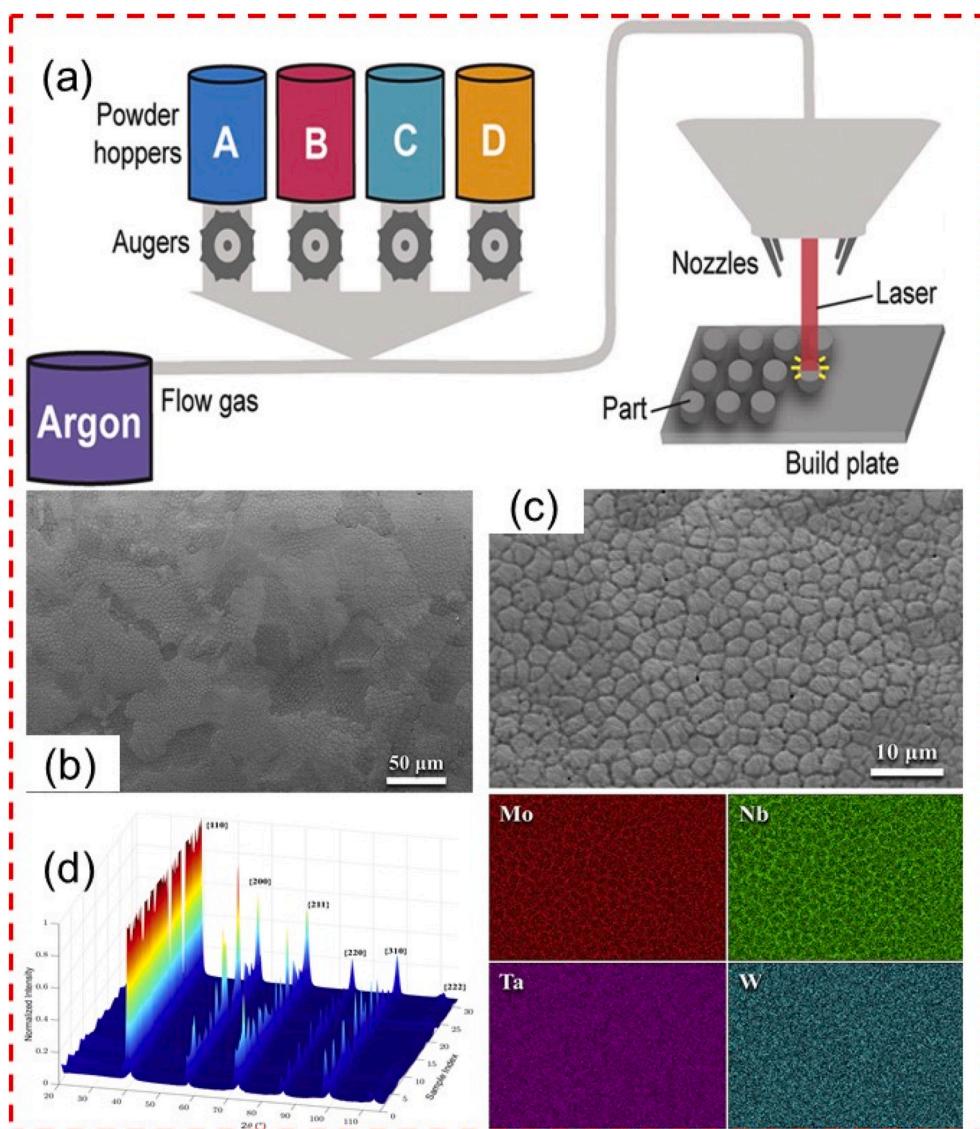


Fig. 7. (a) Schematic illustration of the DED system., (b) SEM image of the polished top surface of additively manufactured nearly equimolar MoNbTaW sample., (c) SEM image and EDS chemical mapping of equimolar MoNbTaW produced through DED, (d) Composite image of XRD patterns from the 31 sample stubs analyzed. All samples appear to exhibit only a single-phase, disordered BCC crystal structure [68].

characteristics of printed RHEA products are significantly influenced by the cooling rate of the DED, which is dependent on the laser power and scanning speed used. Phase change has enough time to occur thanks to a drop-in cooling rate brought on by an increase in laser power. Using mixed elemental powders, Dobbstein et al. investigated pulsed-laser DED processing of the MoNbTaW RHEA [69]. Contrary to conventional techniques that rely on compositional gradients such as combinatorial thin films or diffusion couples, the in situ alloying method utilizing DED empowers researchers to fabricate distinct and varied volumes of targeted alloy compositions. This was demonstrated by the remarkable experimentation conducted by Moorehead et al. [94,95] using Mo–Nb–Ta–W RHEAs shown in Fig. 7. They came to the conclusion that all samples had a single disordered BCC crystal structure and were chemically homogenous, with few inclusions of unmelted powder. The additively made material's cellular structure has a significantly finer length scale than the arc-melted material's dendritic structure, which suggests that the additively manufactured material will cool more quickly. The benefits of DED for HEAs, including outstanding mechanical performance, elemental segregation, and grain refining, are enumerated by Gu et al. [96].

Melia et al. have demonstrated the expeditious screening of RHEAs through the utilization of high-throughput DED processing and characterization [87] and discovered that a large portion of the process space contains faults such as unmelted powders and brittle fracture and cracking. According to Dass and Moridi [97], utilizing DED can result in manufacturing flaws such as lack-of-fusion porosity, keyhole porosity, trapped-gas porosity, oxide inclusions, and microchemical segregation during cooling from the liquid state, just like with all metal-based additive manufacturing techniques. However, Pegues et al. [98] showed that the use of mixed elemental powders during in situ alloying via DED can alter how these defects are formed for a specific window of operation parameters as well as cause new types of defects that are not typically seen in DED using pre-alloyed powders.

DED can manufacture big parts at high deposition rates, making it a feasible option for the production of larger RHEA components. DED can create products with good mechanical properties due to the significant energy input necessary during the deposition process. DED can be used to build gradient structures in RHEAs, which will enhance their mechanical and thermal characteristics. Due to its ability to be used with a variety of feedstock materials, including powders or wires, DED offers

flexibility in material selection. Table 2 summarizes all the AM techniques along with different phases of RHEAs.

One of the critical factors is scanning speed, which influences the rate of cooling of the melt pool and, thus, the resultant microstructure. Higher scanning speeds, due to increased cooling rates, often result in finer microstructures but may also result in increased porosity due to insufficient melting of the powder particles. Lower scanning rates can result in coarser, weaker microstructures. Layer thickness is also an important metric to consider because thicker layers can result in greater temperatures and longer cooling periods, resulting in coarser microstructures with lower strength. Thicker layers, on the other hand, can help reduce processing time and increase productivity. Powder parameters such as particle size distribution, shape, and content can all have a significant impact on the microstructure and properties of RHEAs generated by AM. Powder with a narrow size distribution and a spherical shape creates more homogenous and finer microstructures. Furthermore, utilizing powders with a high degree of alloying can result in higher strength but lower ductility. Finally, the use of process gases like argon or nitrogen can have an effect on the microstructure and characteristics of RHEAs generated by AM. The use of an inert gas can aid in the prevention of oxidation and the reduction of porosity, resulting in better mechanical qualities.

Machining and processing are also big challenges for RHEAs in this decade. Although there is limited research on the cutting performance and machinability of RHEAs, some studies have addressed this issue. For instance, Yang et al. investigated the machining performance of RHEAs, focusing particularly on the WNbMoTaZr_{0.5} alloy [109]. Despite WNbMoTaZr_{0.5} exhibiting the highest cutting forces, its surface roughness was comparatively lower than that of GCr₁₅ and Cr₁₂MoV alloys. They concluded that machining efficiency could be improved by increasing cutting depth without affecting surface roughness. Additionally, they observed that at lower cutting speeds, a brittle fracture phenomenon resulted in frequent chip and workpiece separation, leading to the formation of dense, small-scale patterns. Conversely, with an increase in cutting speed, the material's plasticity improved, resulting in less dense, larger-scale patterns. Achieving better processing performance is crucial for the engineering applications of RHEAs. Wire Electrical Discharge Machining (WEDM), often utilized for processing difficult-to-cut materials like titanium alloys and superalloys, has shown potential for RHEAs [110–112] [110–112] [110–112]. Ceritbinmez et al. conducted experiments on HfTaTiVZr RHEA using WEDM, studying machined surface integrity with different parameter combinations [113]. WEDM performance is influenced by total discharge energy, which can be calculated from single pulse energy and discharge frequency [110–112,114–117]. Additionally, the constituent phases of RHEAs can impact processing performance; for instance, the presence of a second phase in WNbMoTaZr_x RHEAs has been shown to facilitate material removal during WEDM. Therefore, further research on the effect of constituent phases on WEDM performance is necessary for widespread engineering applications of RHEAs.

4. Mechanical properties of RHEAs

RHEAs are expected to have exceptional properties in this area because the melting point of a material is intimately correlated with its transition temperature and diffusion rate for metallic materials. Indeed, RHEAs that include W have remarkable thermal softening resistance. The mechanical characteristics of a finished product are crucial in determining if AM can be used in place of conventional manufacturing procedures in terms of service quality and longevity. The mechanical characteristics of RHEAs are greatly influenced by phases [118,119]. AM, which includes SLM, EBM, and fused extrusion, is a game-changing method for producing near-net-shape metallic components, which is especially critical when producing pricey RHEAs. There are primarily two types of laser energy input: laser metal deposition (LMD) and selective laser melting (SLM) using in-situ alloying of mixed elemental

Table 2

Summary of phases and crystal features observed in various RHEA products developed by DED, SLM, and EBM.

Methods	RHEAs	Phases	Crystal features	Ref.
DED	TiZrNbTa	BCC	From equiaxed grains to elongated grains, along with grain sizes of 2–60 μm	[99]
	TiZrNbHfTa	BCC	Equiaxed grains with random orientations	[100]
	TiZrNbMoV	BCC + NbTi ₄	Dendrites composed of BCC and NbTi ₄ -type phases with Zr-rich precipitates; no dendritic segregation	[101]
	ZrTiVCrFeNi	C14 + αTi	Irregular cell-like structure with an average grain size of 50 μm	[102]
	La _{0.1} Ni _{0.5} Fe _{0.1} V _{0.1} Mn _{0.2}	FCC + La (Ni, Mn) ₅	FCC dendritic microstructure	[103]
	La _{0.03} Ni _{0.17} Fe _{0.2} V _{0.2} Mn _{0.4}	σ + La(Ni, Mn) ₅	Granular La(Ni, Mn) ₅ precipitates; (V, Mn)-rich needle-like precipitates in the σ matrix	[103]
SLM	AlCoCrFeCu/Y ₂ O ₃ /ZrO ₂	I + AlCu ₂ Zr + amorphous	Block-shape crystalline phases composed of ultrafine nanocrystals	[104]
	Co _{1.5} CrFeNi _{1.5} Ti _{0.5} Mo _{0.1}	FCC + single cubic + HCP	Fine uniform microstructure with no visible segregations	[105]
	MoNbTaW	BCC	Lamellar martensite structure dendrites with the primary and secondary dendrite arm spacings of 6.59 and 1.68 μm, respectively	[106]
	Ni ₆ Cr ₄ WFe ₉ Ti	FCC + Unknown phase	Columnar dendritic and fine equiaxed dendritic structures; tiny precipitates (unknown phase) with sizes of 30–50 nm dispersed on intragranular boundaries	[107]
EBM	Co _{1.5} CrFeNi _{1.5} Ti _{0.5} Mo _{0.1}	single cubic and FCC + Ni ₃ Ti	highly columnar prior β grain structure transforming to a bulk microstructure of diffusional α + β.	[108]
	Al _{0.5} CrMoNbTa _{0.5}	BCC + residual porosity	Al-poor phase is based on BCC TaNbMo solid solution	[83]

powder blends or pre-alloyed powders. Zhang et al. [106] can exhibit three performance improvements for NbMoTaW RHEA produced using SLM, shown in Fig. 8. The initial aspect concerns microstructure (Fig. 8 (a)), whereby examination of the microstructure pattern revealed that the HEA sample produced by SLM displayed an exceptionally fine grain and dendritic structure with diameters of 13.4 μm and 6.59 μm , respectively, which is notably smaller than that observed in as-cast samples (Fig. 8(c)). The microhardness of RHEAs is often considered a significant parameter as it is influenced by the BCC phase structure commonly found in most RHEAs. This phase structure contributes to their larger lattice size and increased solid solution hardening compared to alloys with an FCC phase [109]. The microhardness of RHEAs fabricated using laser technology demonstrates a higher value compared to RHEAs fabricated using additive manufacturing and powder metallurgy techniques in most research studies [106,120–129]. Examining Fig. 8(d) reveals the microhardness values of RHEAs manufactured using laser and alternative methodologies. The elevated degree of mixing entropy exhibited by alloys with high-entropy composition imparts upon them a pronounced strengthening effect through the solid solution, resulting in a noteworthy enhancement of their hardness [128].

In this regard, a feasible approach for SLM production of WMoTaTi RHEA involves the use of pre-alloyed powders produced via mechanical alloying and gas-solid fluidization, which yielded a billet with a relative density of 95.8 % and a microstructure consisting of bcc solid solution and hcp-Ti precipitates [83].

In another study, Zhang et al. [112] also studied the addition of TiNi elements in NbMoTa alloy during the SLM process. The addition of Ti decreased the crack size, while the addition of Ni eradicated the microcracks in the NbMoTa RHEA, which exhibits severe cracking flaws during the SLM process. However, the NbMoTaTi alloy showed significant penetrating cracks due to the main phase's declining flexibility (Fig. 9(g)). Fig. 9(c) predicts NbMoTaTi_{0.5}Ni_{0.5} alloy created via SLM; the simultaneous addition of Ni and Ti inhibited the growth of fractures and flaws while also ensuring high room-temperature (2297 MPa) and high-temperature compressive strengths (651 MPa at 1000 °C). Mechanically speaking, adding Ni and Ti increases the NbMoTa alloy's

room-temperature compressive strength by 77.5 %. Additionally, the plasticity and toughness will be improved through the proper heat treatment method and adjustment of the Ti to Ni ratio (Fig. 9(d)).

Fig. 10(e) and (b) show that the NbMoTaTi alloy partially peels off in the direction of the grain boundaries when subjected to tensile stress. After the peeled area, the grain boundary phase (GB phase) is broken, resulting in a smooth fracture. Both the main phase and the GB phase solidify when cracks appear at the grain boundary, solving the issues of sluggish grain development and insufficient liquid filling. The stress and distortion during cooling are overcome by the GB phase of the NbMo-TaTi_{0.5}Ni_{0.5} alloy. The interstitial phase and main phase are effectively integrated, as seen in Fig. 9(g). Additionally, many prolonged dislocations manifest themselves during the GB phase due to tensile stress. Aside from LMD and SLM, EBM is another notable additive manufacturing process for metals and alloys. V.V. Popov et al. [83] used elemental powder mixes to create Al_{0.5}Cr–MoNbTa_{0.5} RHEA via EBM. It is possible to view the pre-heated base plate, pre-sintering, layer-by-layer melting, and ultimate separation from the base plate. Following that, comparable work on investigating the compositional space of W_x(CoCrFeMnNi)_{100-x} RHEAs reported the development of a single phase with W contents less than 6 %, followed by precipitation of Fe₇W₆ above the value and further precipitation of another bcc phase with relatively good efficiency [136]. It was brought to attention that in the absence of pre-alloying, elemental powders featuring low melting points would undergo over-vaporization, leading to incomplete alloying during the process of selective laser melting (SLM). To mitigate this issue, it is imperative to utilize spherical pre-alloyed powders in additive manufacturing techniques to ensure the production of high-quality products.

Another recent study by Gou et al. [119] focuses on the ductile behavior of RHEAs via the AM process, shown in Fig. 10. The stress-strain curves of TiZrHfNb_x RHEAs, printed with varying concentrations of Nb and loaded in different orientations, have been presented in Fig. 10(a) at ambient temperature. Under vertical loading conditions, TiZrHfNb_{0.6} RHEA undergoes fracture prior to yielding, as can be observed in the accompanying insert in Fig. 10(a). TiZrHfNb_{0.8} RHEA,

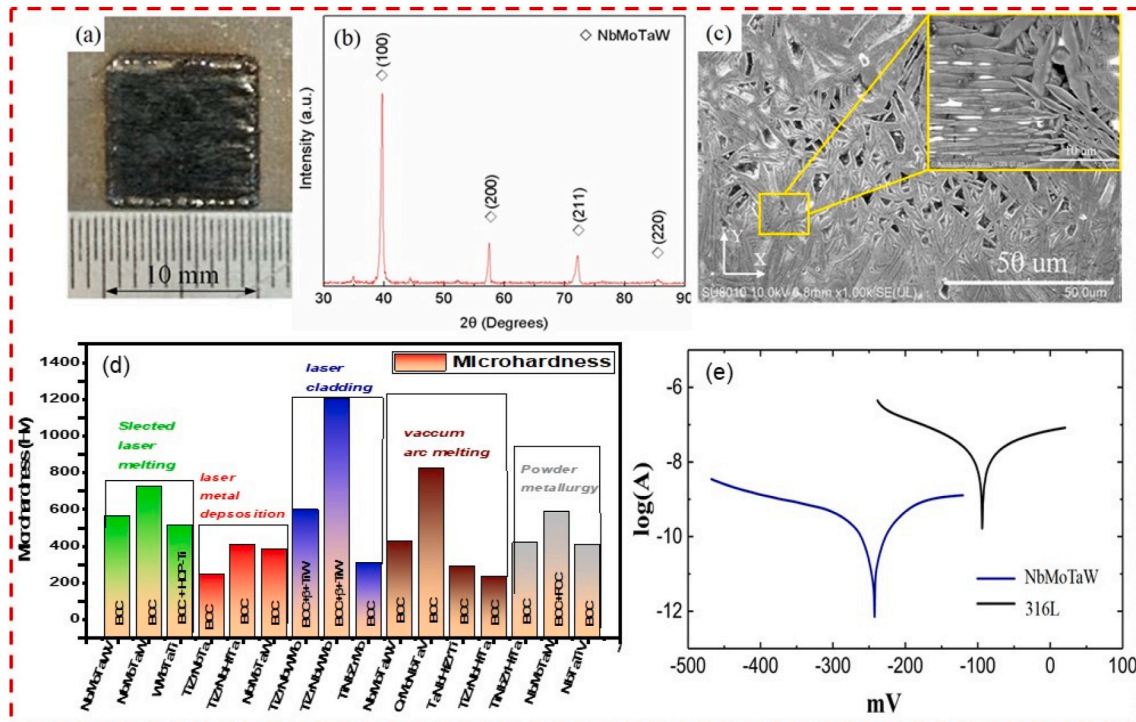


Fig. 8. (a) Patterns of NbMoTaW high-entropy alloy (HEA) parts fabricated by SLM., (b) XRD result with one alloy structure, (c) Pattern grain figure with a further amplification, (d) Microhardness of RHEAs fabricated by different technologies, (e) Polarization curve patterns of WMoNbTa HEA and 316 L [106].

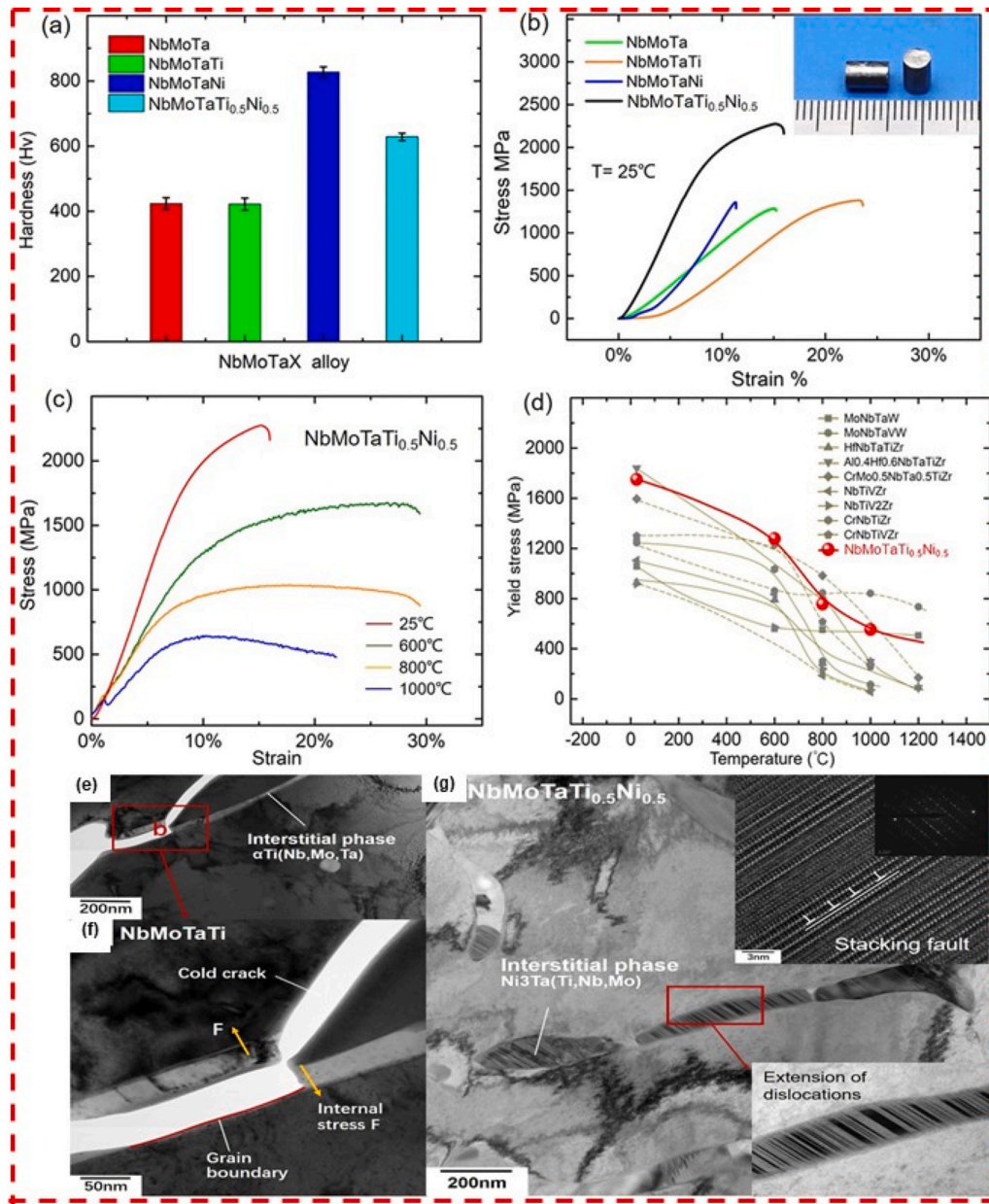


Fig. 9. The NbMoTaX alloy's compressive and hardness characteristics: (a) hardness; (b) compressive strength at room temperature; (c) compressive strength at high temperature; and (d) performance comparison of NbMoTaTi_{0.5}Ni_{0.5} alloy and previously reported high-entropy alloys [130–134]. The microstructural and structural transformations of NbMoTaTi and NbMoTaTi_{0.5}Ni_{0.5} alloys under the influence of tensile stress at grain boundaries: (e) TEM image of the interstitial phase in NbMoTaTi alloy; (f) the fracture of GB phase in NbMoTaTi alloy under stress; and (g) stacking faults in GB phase of NbMoTaTi_{0.5}Ni_{0.5} alloy under stress [135].

on the other hand, exhibits a tensile yield strength of 782 MPa and a ductility of 13.1 %. Finally, the TiZrHfNb variant displays the highest tensile yield strength, amounting to 1048 MPa, and a reasonable ductility of 10.0 %. Of particular note is the superior mechanical performance of TiZrHfNb RHEA when subjected to horizontal loading, displaying a tensile yield strength of 1034 MPa and a ductility of 18.5 %, as depicted in Fig. 10.

The combination of tensile yield strength and elongation for the as-printed TiZrHfNb RHEA is presented in Fig. 10(b) and is compared to Ti alloys, superalloys, and arc-melted RHEAs that were created additively. This comparison serves to illustrate that the TiZrHfNb RHEA printed here possesses outstanding mechanical properties. The fracture surfaces of deformed TiZrHfNb_{0.6} and TiZrHfNb RHEAs are displayed in Fig. 11(c) and (d), respectively. In the TiZrHfNb_{0.6} RHEA following deformation, one can observe intricate fluvial patterns and cleavage

stages, as illustrated in Fig. 10(c). Additionally, an alluring embossment of grains that ingeniously showcases the accumulated deformation at the grain boundaries is evident in Fig. 10(c). The brittle fracture of the printed TiZrHfNb_{0.6} RHEA is perceptible through these aforementioned features. However, a keen observer can discern ductile tearing on the postmortem fracture surface of the TiZrHfNb variation (refer to Fig. 10(d)), along with ductile dimples of various diameters, each smaller than 15 μm , which add to the aesthetic appeal of the image. Based on the EBSD phase maps, the proportion of phase in TiZrHfNb_{0.6} RHEA is estimated to be approximately 15.9 % (see Fig. 10(b)), with just a solitary bcc phase present in the TiZrHfNb version (Fig. 10(c)). Zhang et al. [135] also developed NbMoTaTi_{0.5}Ni_{0.5} RHEA alloys, which are sustainable at high room temperature compressive strength of 2297 MPa and at 1000°C compressive strength of 651 MPa via SLM.

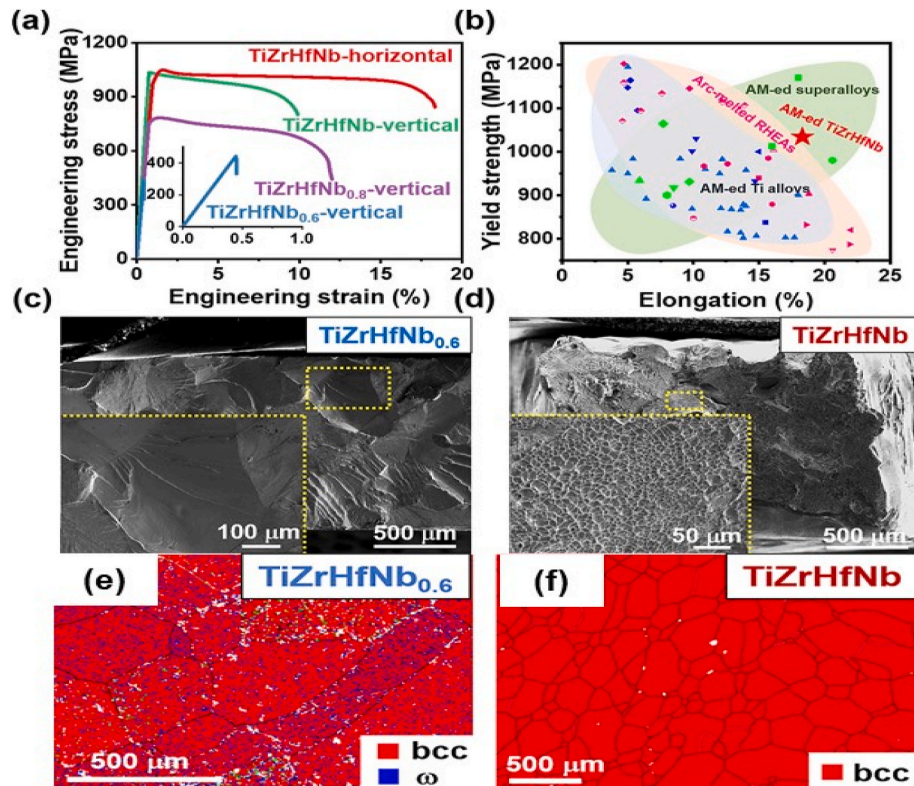


Fig. 10. TiZrHfNb RHEAs printed with different morphologies and mechanical characteristics. (a) Tensile engineering stress-strain curves of room-temperature TiZrHfNb RHEAs with various Nb contents and loading orientations (the inset is the stress-strain curve of TiZrHfNb_{0.6} alloy). Excellent mechanical qualities can be shown in the printed TiZrHfNb RHEA. (b) Yield strength versus elongation of additively made TiZrHfNb RHEA compared to other typical additively manufactured Ti alloys, superalloys, and arc-melted RHEAs [137–140]. (c) Deformed TiZrHfNb_{0.6} and (d) TiZrHfNb RHEA fracture surfaces. The inserts are magnified versions of the dashed line boxes in (c) and (d). The mechanical properties of as-printed RHEAs vary from brittle to ductile fracture as Nb content increases, according to EBSD phase maps of as-printed (e) TiZrHfNb_{0.6} and (f) TiZrHfNb RHEAs [141].

5. Corrosion and oxidation behavior of RHEAs for AM

Refractory High-Entropy Alloys are highly effective in high-temperature settings because of their strong mechanical properties. The mechanical attributes of RHEAs can be enhanced by carefully adjusting their microstructure and composition. When facing harsh environments, it is important for RHEAs to have excellent resistance to corrosion in order to withstand exposure to aggressive substances. Moreover, their ability to maintain strength and thermal stability at high temperatures is essential for ensuring durability and performance in extreme heat. Therefore, tailoring RHEAs for specific applications requires a precise balance of mechanical strength, resistance to corrosion, and resilience to high temperatures. Senkov et al. [134,143] created NbMoTaWV and TaNbHfZrTi RHEAs with bcc structures for high-temperature applications, demonstrating great temperature strength and outstanding phase stability at 1473 K. However, these RHEAs still have the drawback of high density as a result of the existence of Hf and Ta components. Senkov et al. explored CrNbTiVZr HEAs to lower density and increase the oxidation resistance of RHEAs [144]. The researchers made a fascinating discovery regarding Cr-containing HEAs, wherein multiple Laves phases are present, making them brittle at room temperature. Hence, it is of utmost importance to explore novel RHEAs that boast of cost-effectiveness, low density, and high ductility. Conversely, TiZrNbMoV RHEA appears to be the ideal choice for high-temperature engineering and manufacturing industries due to their lighter weight compared to NbMoTaW RHEAs. To investigate the equilibrium bulk characteristics of TiZrNbMoV RHEAs, Tian et al. employed an ab initio technique, revealing promising results [145]. Zhang et al. looked into the influence of V on phase formation and mechanical characteristics [135]. It has been demonstrated that the

alloying effect of the V element in these HEAs can be insignificant. The corrosion resistance of AlCoFeNiTiVZr, AlCoFeNiSm_{0.05}TiV_{0.95}Zr, AlCoFeNiV_{0.9}Sm_{0.1}, and AlCoFeNiSm_{0.1}TiV_{0.9} RHEA products printed by SLM was compared by Saraswat et al. [12]. In order to calculate various parameters, such as corrosion current density (i_{corr}), corrosion potential (E_{corr}), and pitting potential (E_{pit}), standard cathodic and anodic polarization curves were plotted. These values were estimated using the Tafel slope at the appropriate locations on the polarization curves (see Fig. 11). AlCoFeNiSm_{0.05}TiV_{0.95}Zr and AlCoFeNiTiVZr alloys had a pitting potential value of 0.01 V (vs. SCE), but AlCoFeNiSm_{0.1}TiV_{0.9} had a lower value. The pitting potential values of the first two alloys are comparable to those of other HEAs and are higher than those of a number of Al alloys, Cu alloys, and several Ti-based alloys [146]. The corrosion current density in this instance was lowest for the case of AlCoFeNiV_{0.9}Sm_{0.1}, measuring 6 A/cm². Due to their larger corrosion potentials and lower corrosion current densities, Zr-based HEA goods outperformed Zr-free products in terms of corrosion resistance. Additionally, the increased temperature corrosion behavior of the printed AlCoFeNiTiSm_{0.1}V_{0.9} and AlCoFeNiSm_{0.1}V_{0.9} products was observed. By perfecting the alloy's composition and managing the microstructure, RHEAs' corrosion resistance can be raised. For instance, adding metals like Cr and Mo to PBF can considerably increase RHEAs' corrosion resistance [12]. The remarkable ability of these materials to withstand corrosion in a harsh syngas atmosphere at elevated temperatures hints at their capacity to withstand extreme conditions. The corrosion attributes of AlCoCrFeNi HEA products, produced through DED printing and subjected to varying temperatures, were evaluated by Wang et al. [147]. By adjusting their microstructure and composition, for example, by including refractory elements that produce stable carbides and nitrides at high temperatures, RHEAs' high-temperature characteristics can be

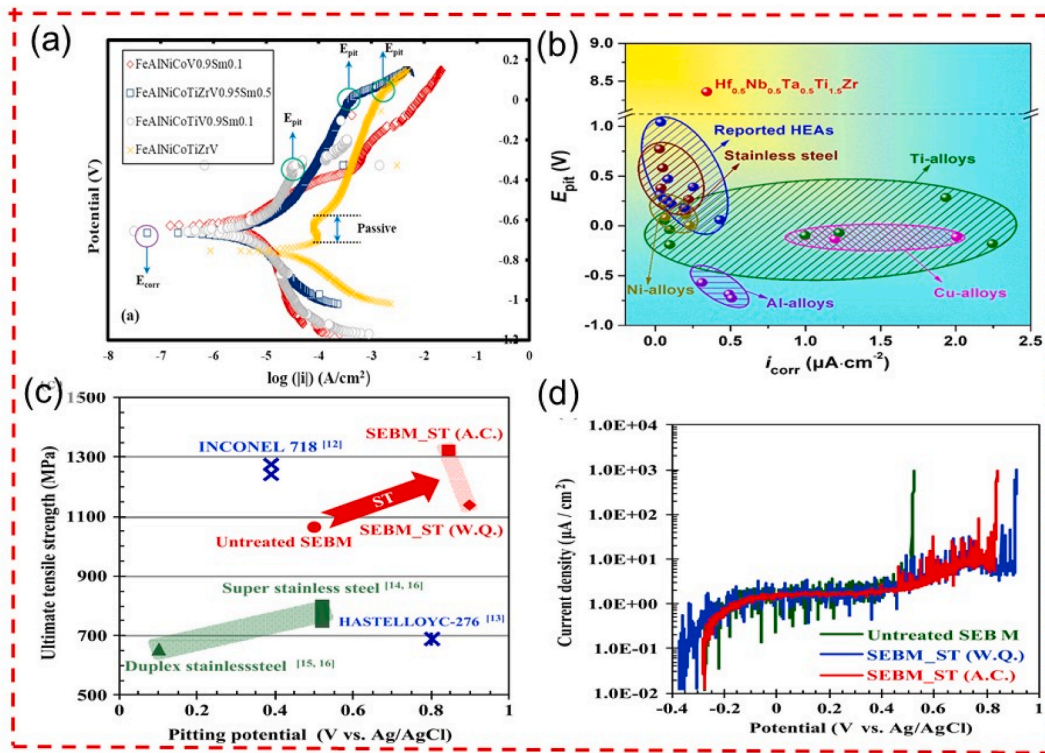


Fig. 11. (a) Potentiodynamic scan plots (potential (V vs. SCE) vs. $\log |i|$) was performed using 3.5 wt% NaCl solution in water [12]. (b) Compare the corrosion of an Hf_{0.5}Nb_{0.5}Ta_{0.5}Ti_{1.5}Zr RHEA with other HEAs and other alloys in a 3.5 wt% NaCl solution [142]. (c) Comparison of tensile strength and pitting potential between untreated SEBM specimens, solution-treated SEBM specimens, and conventional alloys used in severe corrosion environments. (d) Potentiodynamic polarization curves depicting corrosion properties of untreated and solution-treated SEBM specimens in 3.5 % NaCl solution at 353 K [106].

enhanced [148]. According to several studies, AM-fabricated RHEAs offer better high-temperature characteristics than conventionally made alloys [149]. Zhang et al. delved into the intricacies of NbMoTaW and 316 L steel patterns, taking them on a high-voltage ride within the 1.5 to +2 V range. This electrifying experiment took place amidst a 3.5%wt NaCl solution environment [106]. Two types of patterns were corroded in this solution, but the degree of corrosion was quite different; the free-corrosion current for NbMoTaW is $i_{corrH} = 8.716 \times 10^{-11}$ A, and for 316 L steel, $i_{corrS} = 8.815 \times 10^{-9}$ A; the free-corrosion current of RHEA is reduced by two orders of magnitude when compared to 316 L steel. As a result, the NbMoTaW has greater corrosion resistance in general. Zhou et al. [150] concluded from this experiment that this alloy has a greater corrosion resistance than other RHEAs and 316L stainless steel, as shown in Fig. 11. The corrosion resistance of Hf_{0.5}Nb_{0.5}Ta_{0.5}Ti_{1.5}Zr RHEA was tested in a 3.5 wt% NaCl solution. Tantalum and OH⁻ species combine to provide a highly immune passive film [146]. According to Motallebzadeh et al. [151] TiZrTaHfNb and Ti_{1.5}ZrTa_{0.5}Hf_{0.5}Nb_{0.5} have stronger resistance to pitting corrosion and general corrosion than Ti₆AlV, 316L, and CoCrMo alloys and are ideal for biomedical applications. Ti_{1.5}ZrTa_{0.5}Hf_{0.5}Nb_{0.5} alloy was shown to have a strong corrosion resistance when compared to TiZrTaHfNb alloy. The potentiodynamic polarization curves of treated and untreated EBM-printed products are shown in Fig. 11(d). These curves reveal that the passivation state ceased at the pitting potential when a dramatic increase in the corrosion current took place. After the solution treatment, the products' pitting potentials significantly increased. In comparison to traditional alloys utilized in highly corrosive environments, the treated RHEA products had higher strengths as well as higher pitting potentials (Fig. 11(c)).

RHEAs frequently go through a Laves phase, which is correlated with the presence of Cr, Ta, Nb, and other elements. Precipitation of the Laves phase, which often causes a rapid drop in the ductility of RHEAs at room temperature but can significantly boost their high-temperature strength and oxidation resistance, can occur in the BCC matrix. Therefore, it is

crucial to comprehend how the Laves phase is formed. High melting points make refractory metals W, Mo, Ta, and Nb ideal candidates for the RHEAs system. Due to its lower density and the creation of more stable oxides at high temperatures, Nb is more appealing than W, Mo, and Ta [152]. Zr and Ti can be added to the alloy system [144,149,152] to reduce further its density (Zr = 6.51 g/cm³ and Ti = 4.51 g/cm³). In conclusion, AM has created new possibilities for modifying the microstructure and composition of RHEAs in order to adjust their mechanical properties. By adjusting the process variables and alloy composition, RHEAs made by AM can have greater strength, ductility, and fracture toughness. Moreover, RHEAs made using AM technology have better high-temperature and corrosion resistance characteristics, making them promising candidates for usage in severe environments.

The ability of RHEAs to resist oxidation is widely attributed to the creation of protective oxide layers [153]. The incorporation of chromium, aluminum, and silicon was discovered to be efficacious in enhancing the resistance to oxidation of alloys that operate at high temperatures. This improvement is attributed to the formation of protective layers of chromium oxide (Cr₂O₃), aluminum oxide (Al₂O₃), or silicon dioxide (SiO₂) on the surface of the alloys [154]. The elements have a tendency to develop unstructured oxide scales or evaporate as oxides when exposed to higher temperatures in the surrounding air. As a result, oxygen will continuously permeate into the alloy matrix. Taking into account the specific operational conditions, it becomes imperative for RHEAs to possess high resistance to oxidation at elevated temperatures in order to ensure a prolonged lifespan. Chen et al. [155] incorporated TiC into the SLM-fabricated WMoTaNbV material in order to augment its resistance to oxidation. The application of SLM fabrication to the TiC/WMoTaNbV material resulted in the generation of cellular structures and carbides, which serve to facilitate the development of a compact oxide layer. They analyzed WMoTaNbV and TiCWMoTaNbV-based alloys. The influence of SLM process parameters on the surface morphology, internal defects distribution, and

microstructure of the alloy. The TiCWMoTaNbV alloy showed a significant reduction in the number of cracks and improved oxidation resistance. The presence of cellular structures, carbides, and TiO_2 phase contributed to the improved oxidation resistance of the TiCWMoTaNbV alloy shown in Fig. 12. Fig. 13(a) and (b) exhibit the XRD patterns of the surface oxidation products of the TiC/WMoTaNbV-based alloy that underwent oxidation at a temperature of 600 °C for durations of 10 h and 50 h, respectively. Following the oxidation process for 10 h, apart from the oxidation products of $\text{Ta}_{16}\text{W}_{18}\text{O}_{94}$ and $\text{Nb}_{14}\text{W}_3\text{O}_{47}$, the surface of the TiC/WMoTaNbV-based alloy also exhibits the identification of the multi-element oxides of $\text{Ti}_2\text{Nb}_{10}\text{O}_{29}$ (Fig. 12(a)). Furthermore, small amounts of Ta_2O_5 , Nb_2O_5 , MoO_3 , and TiO_2 can be observed. Additionally, after the TiC/WMoTaNbV-based alloy is oxidized for 50 h, the surface primarily consists of multi-element oxides, namely $\text{Ta}_{16}\text{W}_{18}\text{O}_{94}$, $\text{Nb}_{18}\text{V}_4\text{O}_{55}$, $\text{Nb}_{14}\text{W}_3\text{O}_{47}$, and $\text{Ti}_2\text{Nb}_{10}\text{O}_{29}$ (Fig. 12(b)). As demonstrated by the observation of Fig. 12, there has been a notable increase in the burst phenomenon occurring on the surface of all WMoTaNbV samples. Additionally, the oxide scale present on the surface is evidently raised (Fig. 12(a1), (b1), and (c1)). Furthermore, there is a significant augmentation in the quantity of oxide particles found on the surface (Fig. 12 (a3), (b3), and (c3)). Müller et al. investigate the oxidation characteristics of TaMoCrTiAl, NbMoCrTiAl, NbMoCrAl, and TaMoCrAl at a temperature of 1000 °C [154]; it was observed that a protective coating composed of Al_2O_3 , Cr_2O_3 , and CrTaO_4 was formed on these RHEAs. This coating exhibited reduced oxygen diffusion, thereby enhancing the oxidation resistance of the RHEAs. The $\text{NiSi}_{0.5}\text{CrCoMoNb}_{0.75}$ coating, which was produced by liquid casting onto the $\text{Ti}_6\text{Al}_4\text{V}$ alloy, exhibited exceptional resistance to oxidation at high temperatures. In fact, its performance in this regard surpassed that of the $\text{Ti}_6\text{Al}_4\text{V}$ alloy itself [154]. At a temperature of 1000 °C, a number of layered structures consisting of $\text{Ni}_3(\text{VO}_4)_2$, Ni_3TiO_5 , NiCrO_3 , CoCr_2O_4 , and $\text{Nb}_{0.6}\text{Cr}_{0.4}\text{O}_2$ oxide scale were formed in order to impede the

diffusion of oxygen as well as various alloying elements. The role of Ti addition was of great significance in the formation of protective layers and the reduction of less favorable oxides. In an investigation into the impact of Ti, V, and Si content on NbCrMoAl_{0.5} alloys [156], it was observed that the inclusion of Ti and Si elements led to a substantial enhancement in the oxidation resistance of the alloys, whereas the incorporation of V elements had a detrimental effect on the oxidation resistance of the alloys. TiC nanoparticles have the capability to generate TiO_2 in high-temperature oxidizing environments and subsequently restrict the diffusion of oxygen [157].

6. Potential application of RHEAs

A valuable technology for producing RHEA items, AM has recently made strides in powder development, printing methods, and printed product performance, as detailed in the preceding sections. Due to their precise microstructures, printed RHEA items exhibit exceptional qualities as compared to products made using traditional production methods because of the rapid solidification in DED and PBF. The use of 3D printing for HEAs enables the creation of lightweight, customizable, and non-assembly-required goods with a variety of material, design, and manufacturing freedoms. RHEAs, which may be created via additive manufacturing (AM), have the potential to be used in a variety of applications. In this section, we will talk about some of the potential uses for RHEAs made with AM.

The aerospace industry has a lot of promise for AM RHEA goods since there is a growing need for lightweight materials. On the basis of elements with lower densities, such as Al, Mg, Be, and Ti, lightweight RHEAs are still being developed. The majority of the RHEAs are immiscible with other transition metals, including Cr, Fe, and Cu, which frequently combine to create intermetallic compounds, making it challenging to manufacture lightweight RHEAs with a single solid solution

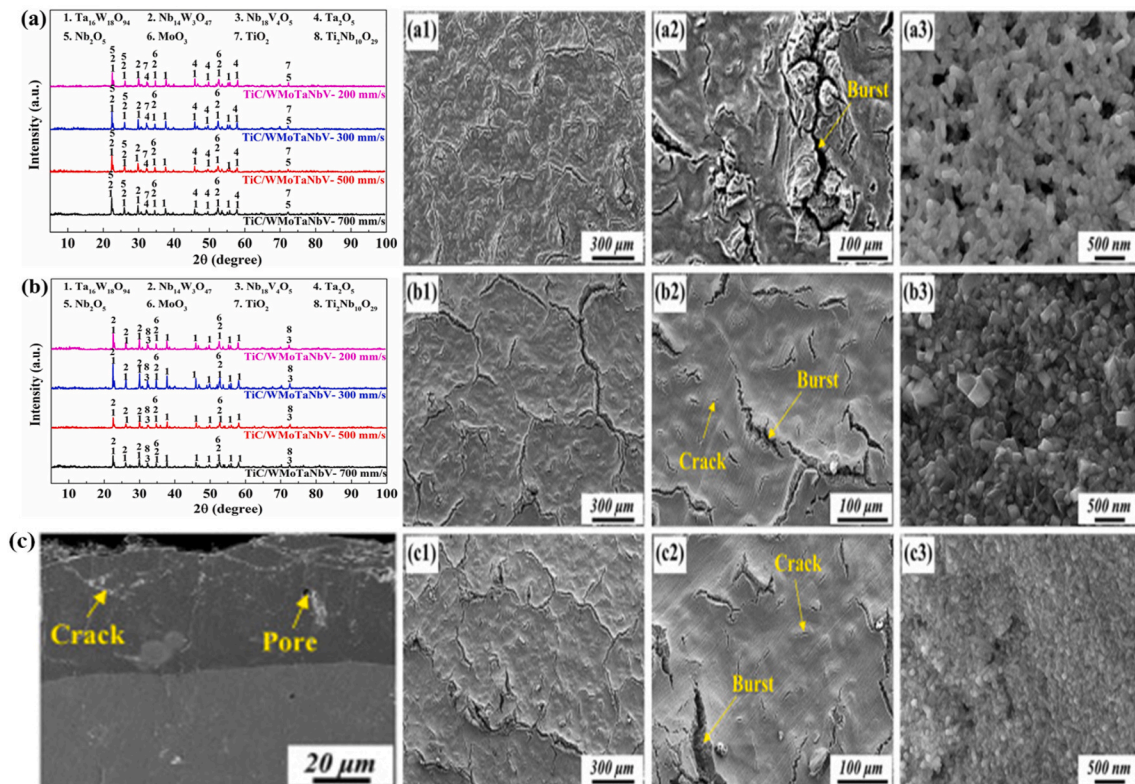


Fig. 12. XRD patterns of the surface oxidation products of TiC/WMoTaNbV alloy oxidized at 600 °C for (a) 10 h and (b) 50 h. (c) Cross-sectional SEM morphologies of WMoTaNbV alloy after oxidation at 600 °C for 50 h. Surface morphologies of the SLMed TiC/WMoTaNbV samples processed at different scanning speeds after high-temperature oxidation at 600 °C for 50 h: (a1), (a2), (a3) 700 mm/s; (b1), (b2), (b3) 500 mm/s; (c1), (c2), (c3) 300 mm/s and (d1), (d2), (d3) 200 mm/s.

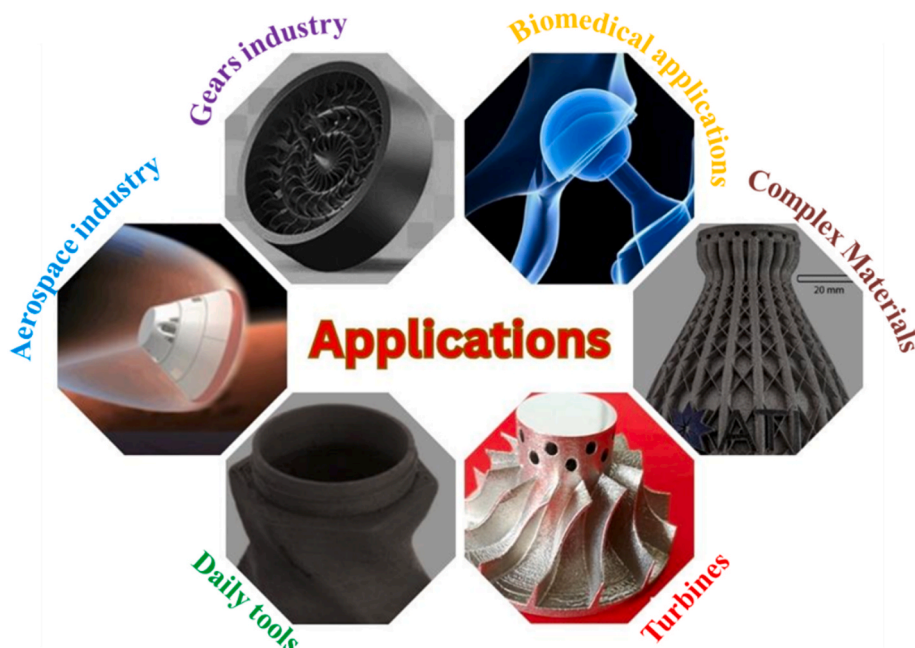


Fig. 13. Potential applications of RHEAs manufactured by AM. Red color for turbines and gears, purple color tools, blue color for complex shapes, yellow color daily life tools, green color for aerospace applications, and light blue for biomedical applications.

phase. In addition to element selection, structural optimization of printed RHEA products can be used to create oxidation-resistant lightweight designs. Materials that can survive harsh conditions, sweltering temperatures, and tremendous strains are necessary for the aerospace and defense industries. RHEAs made with AM have the potential to be used in thermal protection systems, rocket nozzles, and turbine parts, among other things. For instance, a recent study examined the application of RHEAs made by AM in thermal protection systems for hypersonic vehicles. The RHEAs were shown to have high mechanical and thermal stability, making them strong candidates for such applications [158–162] [158–162] [158–162]. Some potential applications are shown in Fig. 13.

Materials that can survive high temperatures and hostile environments are necessary for the energy sector. RHEAs made with AM have the potential to be used in high-temperature fuel cells and gas turbines, as well as other energy-generating and storage systems. As an illustration, a current study examined the usage of RHEAs made by AM as an interconnect substance for solid oxide fuel cells. The RHEAs were found to have excellent high-temperature stability and little thermal expansion, which made them strong candidates for such applications [163, 164]. In the realm of aerospace engineering, RHEAs have been ingeniously utilized as coatings that are both impervious to heat and wear, thereby augmenting the surface attributes of printed objects against the most arduous of environmental circumstances [165]. The nuclear energy sectors use RHEAs as radiation-resistant materials [111,166] [111, 166–168] [111,166–168]. The in-vessel elements of experimental reactors for international thermonuclear usage, consisting of multiple subcomponents, present an opportunity for the creation of three-dimensional printed RHEA merchandise. Through the utilization of stainless-steel powder in the printing of the initial wall panels for one of the reactors, utilizing both SLM and EBM techniques, these reactors provide a prospect for constructing HEA products of high-pressure vessels with superior resistance against irradiation and corrosion. For these components, RHEAs with low thermal neutron absorption, such as TiVCrNbMo [169], TiVZrTa, and TiVCrTa [170] can be suggested. As a result of their high strength, exceptional corrosion resistance, and biocompatibility, RHEAs made with AM have the potential to be used in biomedical applications. For instance, a current study looked into the

usage of RHEAs made by AM for dental implants. RHEAs may also be printed in the AM process for use in the automotive, biomedical, and catalytic industries. RHEAs with features such as TiNbTaZrMo, TiZrHfCrMo, and TiNbTaZrHf that make them potentially suitable for use as orthopedic implants include those with negligible cytotoxicity, high biocompatibility, mechanical, electrochemical, and anti-corrosion capabilities [142,171] [142,171–176] [142,171–176].

In conclusion, the special characteristics of RHEAs made with AM have shown promise for usage in a variety of applications because of their high strength, corrosion resistance, and high-temperature stability. RHEAs created with AM might have a substantial impact on a variety of industries, including aerospace and defense, energy generation and storage, and biomedicine. Beyond the highlighted applications, AM of RHEAs is anticipated to be able to get over production process bottlenecks, which could result in advances in other industries.

7. Conclusion and future prospects of AM with RHEAs

A brand-new class of materials called RHEAs is being created for its high-temperature capabilities, corrosion resistance, and mechanical strength. Explosive progress has been attained in AM for RHEAs, which has been motivated by the promising future of HEAs and the distinctive advantages of 3D printing techniques. In this article, the latest advancements in AM for RHEAs have been meticulously analyzed and evaluated in terms of their powder creation, printing processes, phases, crystal properties, mechanical attributes, functionalities, and potential applications. The production of RHEA powders typically involves sophisticated techniques such as atomization and mechanical alloying. The capability of RHEA powders to be printed majorly depends upon the morphologies, particle size distributions, and element distributions of the powders created. The DED and PBF (SLM and EBM) methods are predominantly employed for printing RHEA products from pre-developed powders as they offer the advantage of quick solidification and the ability to generate intricate geometries. Furthermore, DED can print RHEA products by using unsullied elemental powders from various powder-feeding channels, thus avoiding the time-consuming development of RHEA powders, which is not feasible in PBF.

Despite the rapid cooling rate, unsolicited intermetallic compound

precipitates continue to manifest themselves in the solid solution matrix. To prevent such undesirable precipitates, annealing, solution treatments, and hot isotropic pressing are frequently employed as heat treatments. The higher mechanical qualities of 3D-printed RHEA goods, attributed to their finer grains and simpler phases, surpass those of products made with traditional production methods. By reducing various faults and releasing internal residual stresses, proper heat treatments effectively enhance the qualities of RHEA products, as different printing procedures can produce significant variances in both the strength and plasticity of their products. RHEAs have the potential to emerge in various forms through 3D printing, resulting in devices that are both versatile and weightless, negating the necessity for assembly. Groundbreaking advancements in material and structural optimization have enabled the printing of intricate HEA items, with applications spanning diverse industries such as manufacturing, aerospace, and energy. The potential of additive manufacturing for RHEAs is expounded upon in the following perspectives:

The design of new RHEA powders can consider the inclusion of lanthanide elements (such as Dy, Gd, Lu, Tb, Tm, and others), low-density metal elements (including Be, Li, Mg, Sc, Si, Sn, Zn), precious metal elements (such as Ag, Au, Pd, Pt, Rh, Ru, and more), and high-density metal elements (like Au, Pd, Pt, Rh, Ru, and others). It is crucial, in particular, that the components be robust enough to endure residual stresses and cracks resulting from the rapid solidification of the printing processes. Powder production methods can be optimized to produce innovative RHEA powders with elevated purity, spherical consistency, minimal oxygen levels, and uniform elemental distributions for printing applications.

The capacity to create complicated geometries, lower material waste, and enable the fabrication of single parts in small numbers are just a few benefits that additive manufacturing (AM) techniques provide for the creation of RHEAs. To enhance the quality of printed RHEA products, printing procedures can be developed and improved. In order to reduce cracks and deformation, printing techniques like binder jetting and material extrusion can be used to prevent the thermal stresses present in RHEA materials printed using DED and SLM. For a variety of uses, features of printed items such as oxidation resistance, irradiation resistance, creep resistance, and fracture toughness can be taken advantage of. It is necessary to establish links between material processing and product characteristics so that the underlying mechanisms can be investigated. To provide direction for the solidification behaviors of the RHEAs during 3D printing, RHEA phase diagrams on quaternary or higher systems can be developed. Cellular automata and phase field approaches, respectively, can be used to mimic the microscale crystal development and phase changes in RHEA melt pools. To understand the intricate interaction between the transient thermal history and stress build-up within the goods, thermomechanical models for computing the residual stress existing in printed items can be created using the finite element approach at the macroscale.

Excellent mechanical characteristics of RHEAs made with AM include high strength, ductility, and fracture toughness, in addition to outstanding corrosion resistance and high-temperature qualities. Aerospace and defense, energy production and storage, biological uses, and other future uses are among the applications for RHEAs made with AM. The creation of novel alloys suitable for AM processes, the improvement of AM procedures for RHEAs, and the management of microstructure and flaws are the remaining problems for AM of RHEAs. It is possible to use machine learning to hasten the creation of new RHEAs for 3D printing. For the purpose of the training, a thorough database that includes RHEA data (atomic size difference, mixing entropy, mixing enthalpy, valence electron concentration, electronegativity difference, etc.), printing methods (DED, SLM, EBM, etc.), printing process parameters (laser/electron beam power, scanning speed, hatch space, layer thickness, etc.), and mechanical properties of printed RHEA products (strength, ductility, fatigue life, Based on the database created from the predictions of numerical modeling, experimental printing, and

characterizations, machine learning can be utilized to forecast novel 3D-printed RHEA products quickly.

8. Research gaps and challenges for realizing the full potential of RHEAs

Even though RHEAs can be produced using additive manufacturing (AM), there are still a number of issues that need to be resolved. Some of the unmet needs for AM of RHEAs include:

- Limited options for materials:** The variety of RHEAs that can be manufactured with AM is still constrained when compared to conventional production processes. This is due to the fact that not all RHEAs can be processed using the present AM procedures, necessitating the creation of novel materials and processing methods. Generally speaking, the target alloys' range of applications can be increased by adding new alloying elements to existing alloys, which can produce new alloy types with distinct microstructures and attributes. Mainly composed of Cr, Nb, Mo, Ti, Al, V, Zr, W, and Ta components are the currently produced RHEAs. Elements Ni or Sc have also been added to RHEAs in a number of reports. In light of the general characteristic, low-density metallic elements such as Li, Mg, Y, and light rare-earth (i.e., Cerium group) elements can be included. Furthermore, as previously mentioned by Huang et al. [177], the inclusion of nonmetallic elements like B, C, N, or Si in RHEAs may enhance their mechanical properties.
- Process optimization:** Rapid heat cycling in laser technology provides RHEAs with distinct microscopic characteristics like ultra-fine grains, regular element distribution, and distinctive precipitated phases, which enhances RHEAs' mechanical strength. The variation in the powder element content affects these microscopic features, which manifest as variations in the size and form of the grains and the composition of the precipitation phase. These variations may also have an impact on the properties of RHEAs. However, RHEAs manufactured by SLM and LMD frequently have defects such as unmelted particles, pores, and cracks. In fact, SLM-made RHEAs can even show micro flaws like localized exfoliation at grain boundaries. On the other hand, because LC RHEAs with low coating thickness will generate less heat effect, fewer flaws will develop in LC-fabricated RHEAs. Defects have an impact on RHEA strength and may potentially be the cause of failure in conditions that are corrosive or vibrate. The goal of several investigations has been to reduce the number of defects that are generated in laser-fabricated RHEAs by increasing energy input or adding specific powder constituents. They serve as a guide for enhancing the forming quality of RHEAs produced using laser technologies.
- Microstructural control:** RHEAs with dual- or multi-phase structures have a fatal drawback in that they are relatively weak, particularly in the case of those with brittle precipitates or bcc B2 matrix. Knowing the mechanisms underlying RHEAs' room-temperature toughening would be beneficial. Furthermore, the precipitated Laves phase is widely thought to improve the overall mechanical characteristics of RHEAs; however, specific processes for strengthening and toughening at elevated temperatures are still unknown. Therefore, it is necessary to learn more about the mechanisms of toughening and strengthening in lightweight RHEAs by combining atomistic simulations (such as molecular dynamic simulations) with multi-scale characterization techniques. This will allow for the acquisition of comprehensive mechanical properties.
- Surface quality:** For parts composed of RHEAs created using AM, surface quality is essential. Rough surfaces and residual stresses are examples of inherent characteristics that might alter the mechanical qualities and surface smoothness of RHEAs manufactured by additive manufacturing. Post-processing methods such as machining and polishing are used to improve the quality of the surface. These techniques do have certain disadvantages, though, such they are

costly and time-consuming. Surface polish and mechanical properties of AM-produced RHEAs can be influenced by their uneven surfaces and residual stresses. Common post-processing methods to enhance the surface quality of RHEAs created by AM are machining and polishing. Although machining and polishing can improve surface smoothness, they can be costly and time-consuming processes that raise the cost of manufacturing overall.

- **Scale-up:** RHEA production with AM is still difficult to scale up. Although additive manufacturing (AM) is a versatile manufacturing process that enables the manufacture of small batches of parts, scaling up to large-scale production can be challenging and expensive.

CRediT authorship contribution statement

Muhammad Raies Abdullah: Writing – original draft, Formal analysis, Data curation, Conceptualization. **Zhen Peng:** Project administration, Investigation, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

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